Enhancing EV Charging Performance with ANN-Controlled Solar/Wind Hybrid Systems

CONTENTS

		PageNo
	Abstract	4
CHAPTER I	INTRODUCTION	5-9
1.1	Introduction Project	5
1.2	Literature survey	6-7
CHAPTE II	SYSTEM CONFIGURATION	8-17
3.1	System configuration	8
3.2	DC Microgrid Structure	9
3.3	Power Conversion Units	11
3.4	Charging Units	12
3.5	ANN-Based Controller	13
3.6	Control and Monitoring System	16
CHAPTE IV	CONTROL SCHEME	18-26
4.1	Maximum Power Point Tracking	18
4.2	DC microgrid Configuration	19
4.3	SOC estimation and MSCC algorithm	20
4.4	System modes and control strategy	22
4.5	Multi-stage constant current (MSCC)	23
4.6	ANN	24
CHAPTERV	MODELLING AND SIMULATION	27-30
5.1	Simulation	27
CHAPTE VI	RESULTA AND ANALYSIS	31- 37
6.1	Battery results with PI controlled method (CC)	31
6.2	Battery results with PI controlled method (MSCC)	33
6.3	Battery results with ANN controlled method (MSCC)	35
	Conclusion	38
	Future Scope	39
	Applications	40

References **41**

ABSTRACT

This project explores the enhancement of electric vehicle (EV) charging performance through the integration of an Artificial Neural Network (ANN) controller within a solar/wind hybrid power system. The study focuses on designing a robust, efficient system that leverages the complementary nature of solar and wind energy sources to ensure a consistent and reliable power supply. By incorporating ANN control strategies, the system dynamically adjusts to fluctuations in energy production and demand, optimizing the charging process for EVs. The proposed hybrid system not only improves energy efficiency but also reduces dependency on the conventional grid, promoting sustainable and eco-friendly transportation solutions. Simulation results demonstrate significant performance gains in terms of energy utilization and charging speed, highlighting the potential of ANN-controlled hybrid systems in advancing the EV charging infrastructure.

Keywords: Wind turbine, Solar photovoltaic system, Electric vehicles. ANN controller

CHAPTER-I

INTRODUCTION

Using an electric vehicle can minimize the damages that the transportation section has on the environment. The cost, charging structure, power electrical system, and battery capacity are all factors that affect how well EVs work . The utilization of ecologically friendly energy sources like solar and wind energy is inevitable in light of global warming and the potential for fossil fuel shortages. Utilizing solar and wind energy has the benefit of being cheap, plentiful, pollution-free, and available everywhere. The battery has to be charged in order for the car to operate. No need to mention that the car requires more charge while traveling over long distances. As a result, charging facilities for EVs are situated along the sides of the highway. These automobiles are charged with the help of power produced by renewable energy sources, which are close to the charging station . Furthermore, the energy from various input sources had to be transferred via power converters, including isolated and non-isolated types. Isolated converters are more sophisticated than non-isolated converters due to the use of transformers. As a result, transformers are not required in the circuits of the majority of applications, which instead require a multiport dc-dc converter . In addition, in order to extract maximum power from the solar part of the system, there is a need for a tracking technique. In this paper, the P&O MPPT technique is used to extract the most power possible from the PV component . The main contribution of this paper is to charge EVs with renewable energies both in the daytime and night time. This is how the paper will be presented: The characteristics of the solar/wind system are presented in the next part. The suggested topology is described in part 3. The simulation results are shown in part 4. And finally, the conclusion is in the last part.

LITERATURE REVIEW:

[1] H. S. Das, M. M. Rahman, S. Li, and C. Tan Are proposed in this system Transportation electrification is one of the main research areas for the past decade. Electric vehicles (EVs) are taking over the market share of conventional internal combustion engine vehicles. The increasing popularity of EVs results in higher number of charging stations, which have significant effects on the electricity grid. Different charging strat2egies, as well as grid integration methods, are being developed to minimize the adverse effects of EV charging and to strengthen the benefits of EV grid integration. In this paper, a comprehensive review of the current situation of the EV market, standards, charging infrastructure, and the impact of EV charging on the grid is presented. The paper introduces the current EV status, and provides a comprehensive review on important international EV charging and grid interconnection standards. Different infrastructure configurations in terms of control and communication architectures for EV charging are studied and evaluated. The electric power market is studied by considering the participation roles of EV aggregators and individual EV owners, and different optimization and game based algorithms for EV grid integration management are reviewed. The paper specially presents an evaluation on how the future EV development, such as connected vehicles, autonomous driving, and shared mobility, would affect EV grid integration as well as the development of the power grid moves toward future energy Internet and how EVs would affect and benefit the development of the future energy Internet. Finally, the challenges and suggestions for the future development of the EV charging and grid integration infrastructure are evaluated and summarized.

[2] O. Ekren, C. H. Canbaz, and Ç. B. Güvel, are proposed in this paper Although, fossil fuels are still the main source that used to balance the global demand/supply dynamics of energy; most countries have embarked on a quest to reach a solution and started changing their direction from fossil resources (especially oil and gas) to other resources by creating new opportunities with a focus meeting the increasing energy demand by reducing it via energy efficiency. “Transportation” is one of the biggest items of global energy consumption with 24% usage percentage and most of this amount is supplied by conventional energy resources. However, worldwide supporting policies and regulations are positively affected the starting period of environment-friendly use of energy resources. It expedited the process of new technological developments that minimize waste, lower the level of air pollution caused by fossil fuel-powered internal combustion engines, conserve forests and decrease the emissions of greenhouse gases. Worldwide extension of “Electric Vehicle Development and Production”

is one of the main concern of these regulations. However, increasing the number of electric vehicles also brings sustainability problems such as; supplying the electricity from renewable resources in an efficient and sustainable way, handling the electricity load on the grid, and establishment of new charging stations. Therefore, in this study a wind solar hybrid energy charging station designed and optimized via HOMER software. The sizing methodology is suitable to apply anywhere around the worldwide. The optimal solution for the hybrid system consists of 44.4% wind energy and 55.6% solar energy and the annual electricity production is 843150 kWh with the 0.064 $/kWh production cost.

[3] M. S. H. Lipu et al Electric vehicles are receiving widespread attention around the world due to their improved performance and zero carbon emissions. The effectiveness of electric vehicles depends on proper interfacing between energy storage systems and power electronics converters. However, the power delivered by energy storage systems illustrates unstable, unregulated and substantial voltage drops. To overcome these limitations, electric vehicle converters, controllers and modulation schemes are necessary to achieve a secured and reliable power transfer from energy storage systems to the electric motor. Nonetheless, electric vehicle converters and controllers have shortcomings including a large number of components, high current stress, high switching loss, slow dynamic response and computational complexity. Therefore, this review presents a detailed investigation of different electric vehicle converters highlighting topology, features, components, operation, strengths and weaknesses. Moreover, this review explores the various types of electric vehicle converter controllers and modulation techniques concerning functional capabilities, operation, benefits and drawbacks. Besides, the significance of optimization algorithms in electric vehicle converters is illustrated along with their objective functions, executions and various factors. Furthermore, this review explores the key issues and challenges of electric vehicle converters, controllers and optimizations to identify future research gaps. Finally, important and specific suggestions are delivered toward the development of an efficient converter for future sustainable electric vehicle applications.

[4] G. Kumar, is proposed in this paper recent years, Hybrid Wind-Solar Energy Systems (HWSES) comprised of Photovoltaic (PV) and wind turbines have been utilized to reduce the intermittent issue of renewable energy generation units. The proposed research work provides optimized modelling and control strategies for a grid-connected HWSES. To enhance the efficiency of the maximum power tracking of a grid-connected wind-driven Doubly Fed Induction Generator (DFIG) integrated with solar Photovoltaic (PV) system, connected to the DC link of the back-to-back converters of the Hybrid Wind-Solar Energy System (HWSES).

Stator Flux-Oriented control is utilized to regulate the Grid Side Converter and Rotor Side Converter. The main objective of this paper is to apply the Maximum Power Point Tracking (MPPT) strategy to wind and solar PV systems to maximize the power extraction and to provide better integration of the hybrid systems into the electrical grids. Perturb and Observe (P&O) and Incremental Conductance (IC) MPPT algorithms are implemented to the solar PV system with varying solar insolation and their performances and efficiencies are compared. For varying wind speeds, Tip Speed Ratio (TSR) and Optimal Torque (OT) MPPT algorithms are implemented and their performances and efficiencies are compared for the hybrid system considering and integrating solar PV system. The optimal torque MPPT algorithm shows better responses when compared to the TSR method. A 2MW simulation model of the HWSES is developed and its performance is analysed using MATLAB/Simulink environment. The implemented schemes have the advantage of tracking the optimal power output of the HWSES rapidly and precisely. Additionally, the provided schemes effectively control the power flowing through the HWSES and the utility grid, resulting in a quick transient response and enhanced stability performance.

[5] V. Khare, S. Nema, and P. Baredar, are proposed in this paper The demand for electricity is increasing day by day, which cannot be fulfilled by non-renewable energy sources alone. Renewable energy sources such as solar and wind are omnipresent and environmental friendly. The renewable emulnergy sources are emerging options to fulfill the energy demand, but unreliable due to the stochastic nature of their occurrence. Hybrid renewable energy system (HRES) combines two or more renewable energy sources like wind turbine and solar system. The objective of this paper is to present a comprehensive review of various aspects of HRES. This paper discusses prefeasibility analysis, optimum sizing, modeling, control aspects and reliability issues. The application of evolutionary technique and game theory in hybrid renewable energy is also presented in this paper.

[6] L. Khalvati and L. Khalvati, are proposed in this paper These days, the main issues are fossil fuels and air pollution, which has led several nations to expand their investments in the development of renewable energy. Nowadays, the use of solar energy is growing steadily in popularity across all applications, from residential to commercial. The tilt angle of the solar system is the most important part of the design which is related to the rooftop architecture. The performance of three different architecture rooftop designs is examined in this research, including Flat roof, Gable roof, and Shed roof with the same total area. According to the

modeling results, the shed roof design is the best design with the highest extracted energy and a large amount of air pollutions will be removed from the air.

[7] A. Balal and M. Giesselmann, are proposed in This paper makes use of electric vehicles (EVs) that are simultaneously connected to the Photovoltaic Cells (PV) and the power grid. In micro-grids, batteries of the electric vehicles (EVs) used as a source of power to feed the power grid in the peak demands of electricity. EVs can help regulation of the power grid by storing excess solar energy and returning it to the grid during high demand hours. This paper proposes a new architecture of micro-grids by using a rooftop solar system, Battery Electric Vehicles (BEVs), grid connected inverters, a boost converter, a bidirectional half-bridge converter, output filter, including L, LC, or LCL, and transformers. The main parts of this micro-grid are illustrated and modeled, as well as a simulation of their operation. In addition, simulation results explore the charging and discharging scenarios of the BEVs.

[8] Y. Wu, Z. Liu, J. Liu, H. Xiao, R. Liu, and L. Zhang, are proposed Zinc–air batteries deliver great potential as emerging energy storage systems but suffer from sluggish kinetics of the cathode oxygen redox reactions that render unsatisfactory cycling lifespan. The exploration on bifunctional electrocatalysts for oxygen reduction and evolution constitutes a key solution, where rational design strategies to integrate various active sites into a high-performance air cathode remain insufficient. Herein, a multiscale construction strategy is proposed to rationally direct the fabrication of bifunctional oxygen electrocatalysts for long-lifespan rechargeable zinc–air batteries. NiFe layered double hydroxides and cobalt coordinated framework porphyrin are selected as the active sites considering their high intrinsic activity at the molecular level, and the active sites are successively integrated on three-dimensional conductive scaffolds at mesoscale to strengthen ion transportation. Consequently, the multiscale constructed electrocatalyst exhibits excellent bifunctional performance (ΔE = 0.68 V), which is even better than that of the noble metal based benchmarks. The corresponding air cathodes endow zinc–air batteries with a reduced voltage gap of 0.74 V, a high power density of 185.0 mW cm^{-2}, and an ultralong lifespan of more than 2400 cycles at 5.0 mA cm^{-2}. This work demonstrates a feasible strategy to rationally integrate various active sites to construct multifunctional electrocatalysts for energy-related processes.

CHAPTER-II

ELECTRIC VEHICLES

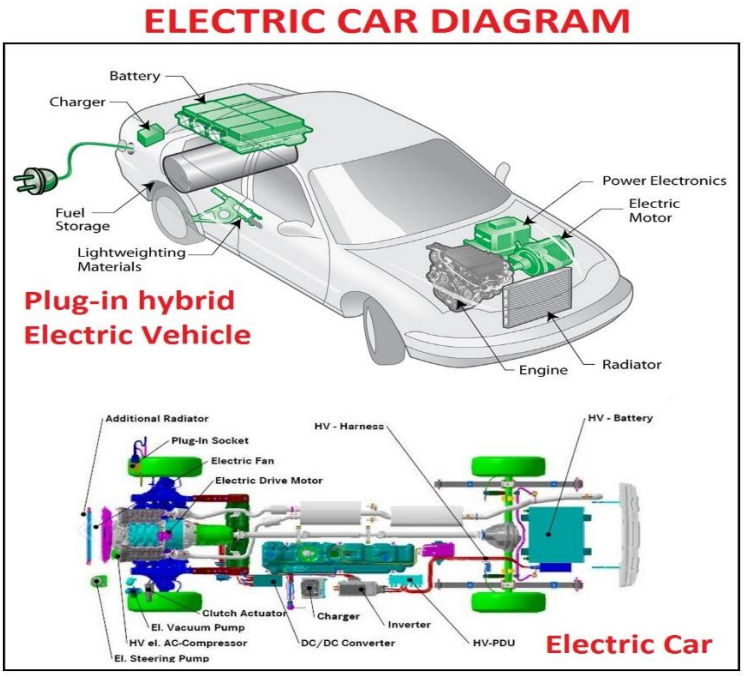


Fig. Electric Vehicle

INTRODUCTION

Electric vehicles (EVs) have a surprisingly long history that dates back to the early 19th century. The first electric vehicle was invented by Scottish inventor Robert Anderson in 1832, which used non-rechargeable primary cells. However, the first practical electric vehicle was not developed until the late 1800s.

In 1884, Thomas Parker, a British inventor, built the first practical electric car, which used lead-acid batteries and had a range of about 48 kilometers. By the turn of the century, electric cars were becoming increasingly popular, particularly in urban areas, as they were quiet, easy to operate, and did not produce exhaust emissions.

However, the invention of the gasoline-powered internal combustion engine and the mass production of automobiles by companies such as Ford quickly made gasoline-powered cars more affordable and popular. As a result, the popularity of electric cars declined, and they became a niche product.

In the 1970s, the oil crisis led to renewed interest in electric vehicles as a way to reduce reliance on imported oil. This led to the development of several new electric cars, including the General Motors EV1, which was introduced in the 1990s.

In the 2000s, advances in battery technology and the need to reduce greenhouse gas emissions from transportation led to a renewed interest in electric vehicles. Companies such as Tesla, Nissan, and General Motors began to produce modern electric cars with longer ranges and better performance.

Today, Electric cars are gaining more popularity and becoming more widespread, as numerous nations worldwide establish goals to eliminate petrol-fueled vehicles in the forthcoming years. The history of electric vehicles shows that while their popularity has waxed and waned over the years, they have the potential to be an important part of the transition to a more sustainable and low-carbon transportation system

What is the need of EVs?

Electric vehicles (EVs) are needed for several reasons:

Environmental concerns: EVs have the potential to reduce greenhouse gas emissions and improve air quality. They produce zero emissions at the tailpipe, which can help reduce the harmful effects of air pollution on human health and the environment.

Energy security: EVs can help reduce a country's dependence on foreign oil and increase energy security by using domestically-produced electricity.

Economic benefits: The production of EVs can create jobs in the manufacturing and supply chain sectors. Additionally, EVs have lower operating costs than traditional gasoline-powered vehicles, which can save consumers money in the long run.

Technological innovation: The development of EVs has spurred technological innovation in battery technology, charging infrastructure, and other related fields. This innovation can lead to new and improved products and services that benefit society as a whole.

Overall, EVs are an important tool for addressing climate change, improving air quality, enhancing energy security, and promoting economic development and technological innovation.

Working of EVs

Electric vehicles (EVs) operate using an electric motor and a rechargeable battery instead of an internal combustion engine. The battery stores electrical energy that powers the motor, which turns the wheels and propels the vehicle forward.

The process of powering an EV starts with plugging the vehicle into an electrical outlet or a charging station. The battery is charged using electricity from the grid, which is converted into direct current (DC) power and stored in the battery.

When the driver wants to move the vehicle, they activate the accelerator pedal. This sends a signal to the controller, which determines how much power to send from the battery to the motor. The motor then converts the electrical energy from the battery into mechanical energy, which turns the wheels and moves the vehicle forward.

As the vehicle moves, the battery gradually discharges, and the range of the vehicle decreases. When the battery level drops below a certain point, the driver needs to recharge the battery using a charging station or electrical outlet.

Regenerative braking is another important feature of many EVs. When the driver brakes, the motor acts as a generator and converts some of the vehicle's kinetic energy back into electrical energy, which is then used to recharge the battery. This helps to extend the range of the vehicle and improve overall energy efficiency.

Types of EVs

Electric vehicles (EVs) are vehicles that are powered by electric motors and batteries rather than by internal combustion engines that burn fossil fuels. There are several different types of EVs, including:

Battery Electric Vehicles (BEVs): These vehicles are powered solely by electricity and have no internal combustion engine. They are charged by plugging them into an external power source, such as an EV charging station, and can travel for hundreds of kilometers on a single charge.

Hybrid Electric Vehicles (HEVs): These automobiles are equipped with both an electric motor and an internal combustion engine that runs on gasoline. The electric motor provides assistance to the engine during acceleration and at lower speeds, resulting in improved fuel efficiency and reduced emissions.

Plug-in Hybrid Electric Vehicles (PHEVs): These vehicles are similar to HEVs, but they have larger batteries that can be charged by plugging them into an external power source. They can travel for a short distance on electric power alone before the engine takes over.

Fuel Cell Electric Vehicles (FCEVs): These automobiles employ hydrogen fuel cells to generate electricity that drives their electric motors. They produce no emissions other than water vapor.

Electric vehicles offer several advantages over traditional gasoline-powered vehicles, including:

Zero tailpipe emissions: EVs produce no exhaust emissions, making them much cleaner and less polluting than gasoline-powered vehicles.

Lower operating costs: Electric motors are more efficient than internal combustion engines, meaning that EVs are cheaper to operate and maintain.

Reduced dependence on oil: Electric vehicles can be powered by a variety of sources, including renewable energy sources like solar and wind power, reducing dependence on oil and other fossil fuels.

Quiet and smooth: Electric motors produce very little noise and vibration, making EVs quieter and smoother to drive than traditional vehicles.

Performance benefits: Electric motors provide instant torque, meaning that EVs can be very quick and responsive, and can offer a smooth and enjoyable driving experience.

Overall, electric vehicles are becoming an increasingly popular and important part of the transition to a more sustainable and low-carbon transportation system

ADVANTAGES

Electric vehicles (EVs) offer several advantages over traditional gasoline-powered vehicles, including:

Zero tailpipe emissions: EVs produce no exhaust emissions, making them much cleaner and less polluting than gasoline-powered vehicles. This can help to improve air quality and reduce the impact of transportation on public health and the environment.

Lower operating costs: Electric motors are more efficient than internal combustion engines, meaning that EVs are cheaper to operate and maintain. EVs also have fewer moving parts, which can reduce the need for maintenance and repairs.

Reduced dependence on oil: Electric vehicles can be powered by a variety of sources, including RES like solar and wind power, reducing dependence on oil and other fossil fuels. This can help to improve energy security and reduce the risk of price volatility associated with oil.

Quiet and smooth: Electric motors produce very little noise and vibration, making EVs quieter and smoother to drive than traditional vehicles. This can provide a more enjoyable driving experience and reduce noise pollution in urban areas.

Performance benefits: Electric motors provide instant torque, meaning that EVs can be very quick and responsive, and can offer a smooth and enjoyable driving experience. EVs can also be designed with a low center of gravity, which can improve handling and stability.

Cost savings: In many cases, EVs can provide significant cost savings over the lifetime of the vehicle. Although they may have a higher purchase price.

APPLICATIONS

Electric vehicles (EVs) have several applications and benefits, including:

Personal transportation: Electric cars, trucks, and motorcycles are increasingly being used as a primary means of transportation for individuals and families. EVs offer a more sustainable and environmentally friendly option compared to traditional gasoline-powered vehicles.

Public transportation: Many cities are adopting electric buses and trains as a way to reduce emissions and improve air quality. EVs can also provide a quieter and more comfortable ride for passengers.

Commercial transportation: Electric trucks and vans are becoming more common in commercial fleets, especially for last-mile delivery and urban transportation. Electric vehicles can offer lower operating costs and reduced maintenance needs, making them an attractive option for businesses.

Renewable energy integration: EVs can serve as a means of storing excess RES like wind and solar power. This helps to balance the grid and ensure a reliable supply of energy.

Emergency services: Electric vehicles are being used by emergency services like police, fire, and ambulance services. EVs offer fast and quiet response times, making them well-suited for urban environments.

Agriculture and mining: Electric tractors and mining equipment are becoming more common in these industries, offering a quieter and more sustainable option compared to traditional diesel-powered equipment.

Overall, the applications of EVs are diverse and offer a range of benefits, including reduced emissions, lower operating costs, and improved energy security.

CHAPTER-III

SOLAR PV GENERATION

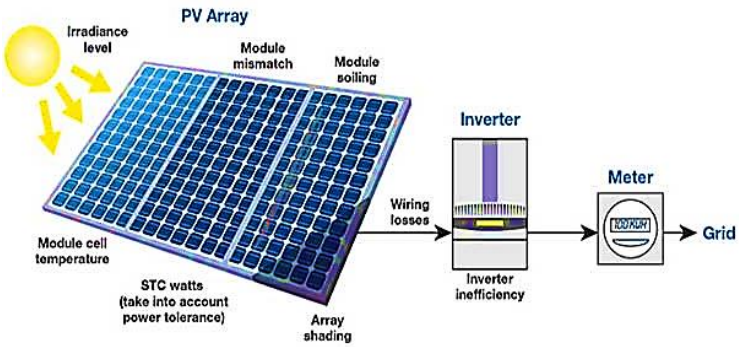


Fig. solar PV generation

Introduction

Solar PV generation is the process of generating electricity from sunlight through the use of solar PV technology. Solar PV systems transform solar energy into DC electricity, which can subsequently be converted into AC electricity that is suitable for powering residential and commercial buildings, as well as other electrical loads.

A solar PV system comprises fundamental components such as solar panels, an inverter, and a mounting system. Solar panels contain numerous solar cells that transform sunlight into DC electricity. The inverter converts DC into AC, which can be utilized to power electrical loads or supplied back to the grid. The mounting system is responsible for securing the solar panels in position and ensuring they are oriented towards the sun for optimal energy generation.

Solar PV generation has become an increasingly important source of renewable energy, as the cost of solar PV systems has decreased over time and their efficiency has increased.

Solar PV systems can be installed on rooftops or on the ground, and can range in size from small residential systems to large-scale utility systems that generate megawatts of electricity.

One of the main advantages of solar PV generation is that it produces electricity without emitting greenhouse gases or other pollutants, making it a clean and sustainable source of energy. Additionally, solar PV systems can provide power in remote locations where access to electricity is limited or expensive, and can reduce reliance on fossil fuels and the grid.

Photo-Voltaic (PV) Cell, Module and Array:

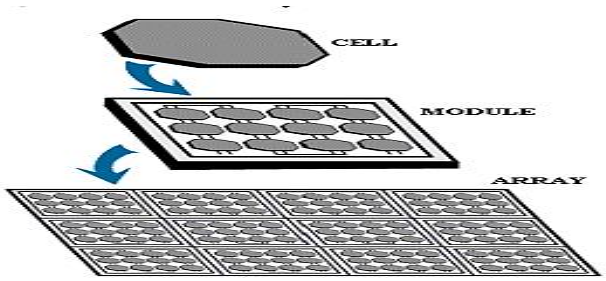


Fig. Cell Module and Array

PV Cell

A PV cell, commonly known as a solar cell, is a device that harnesses the energy from light and converts it into electricity using the PV effect. These cells are usually made of semiconductor materials like silicon and form the crucial building blocks of solar PV systems.

The photovoltaic effect takes place when photons of light energy are absorbed by the semiconductor material in a PV cell. This energy is then transmitted to electrons in the material, stimulating them and producing a flow of electric current. The current flow is subsequently gathered by metal contacts located on the cell and can be utilized as electrical energy.

The efficiency of a PV cell is determined by several factors, including the type of semiconductor material used, the thickness of the cell, and the manufacturing process. Most commercial PV cells have an efficiency ranging from 15% to 20%, with some advanced cells achieving efficiencies of over 25%.

PV cells can be used in a variety of applications, from small, portable devices like calculators and watches to large-scale solar PV systems that can power entire buildings or communities. PV cells are often combined into modules or arrays to provide the desired power output, and the arrays can be mounted on rooftops, on the ground, or in other locations where they can capture sunlight.

Overall, PV cells are a critical component of the solar PV industry, and their development and improvement is essential for expanding the use of solar energy as a renewable and sustainable source of electricity.

PV Module:

A PV module, also known as a solar panel, is a collection of interconnected PV cells that convert sunlight into electrical energy. The PV cells are typically made of silicon or other semiconductor materials and are housed in a protective casing with a glass cover to protect them from the elements.

A PV module is the basic building block of a solar PV system and is designed to produce a specific amount of electrical power. The power output number and size of the PV cells it contains, as well as the efficiency of the cells and the quality of the manufacturing process.

PV modules are designed to be connected together in a series or parallel configuration to create a solar PV array that can produce the desired amount of power. The array can be installed on a rooftop or on the ground, and can be used to power homes, businesses, and other electrical loads.

The efficiency of a PV module is an important factor in its performance, and manufacturers are constantly working to improve the efficiency of their products. Most commercial PV modules have an efficiency ranging from 15% to 20%, with some advanced modules achieving efficiencies of over 25%.

Overall, PV modules are a critical component of the solar PV industry, and their development and improvement is essential for expanding the use of solar energy as a renewable and sustainable source of electricity.

PV Array:

A PV array is a collection of interconnected PV modules that are designed to generate electrical power from sunlight. The PV modules are typically mounted on a support structure, such as a rooftop or ground-mounted system, and are connected together in series and/or parallel to produce the desired electrical output.

The size of a PV array depends on the amount of electrical power that is required and the available space for installation. PV arrays can range from small residential systems that generate a few kilowatts of power to large utility-scale systems that generate megawatts of power.

PV arrays are often used in combination with other components, such as inverters and batteries, to create a complete solar PV system. Inverters are employed to convert the DC power generated by the PV panel into AC power that is suitable for powering electrical loads or can be supplied back to the grid. Excess energy generated by the PV array can be stored using batteries for utilization during periods when the sun is not available, such as cloudy weather or nighttime.

The performance of a PV array is affected by a number of factors, including the amount of sunlight available, the angle and orientation of the array, and the temperature of the PV modules. Proper design, installation, and maintenance of a PV array is essential to ensure that it operates efficiently and reliably over its lifetime.

Overall, PV arrays are a critical component of the solar PV industry, and their development and improvement is essential for expanding the use of solar energy as a renewable and sustainable source of electricity.

Working of PV cell:

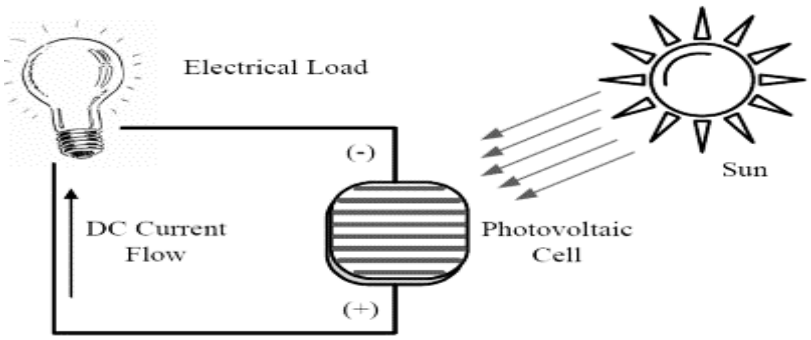


Fig. working of PV Cell

A solar cell, also known as a PV cell, converts light energy into electric energy using the photovoltaic effect. This effect occurs as photons are absorbed by a semiconductor material, typically silicon, resulting in the generation of electric energy.

A solar cell is composed of a pair of slim layers of semiconductor material, with one layer functioning as n-type and the other as p-type. The n-type material has a surplus of available electrons, while the p-type material has a shortage of electrons, or "holes." As light energy shines onto the PV cell, the electrons in the n-type material are stimulated and migrate to the p-type material, resulting in the creation of an electric current within the cell that can be collected through metal contacts.

The electricity generated by the PV cell is in the form of direct current (DC), which can be utilized to power electrical loads, or it can be transformed into alternating current (AC) electricity through an inverter to power homes, businesses, and other electrical devices.

The efficiency of a PV cell is affected by several factors, including the type and quality of the semiconductor material, the thickness of the cell, and the quality of the manufacturing process. Most commercial PV cells have an efficiency ranging from 15% to 20%, with some advanced cells achieving efficiencies of over 25%.

Overall, PV cells are a critical component of the solar PV industry, and their development and improvement is essential for expanding the use of solar energy as a renewable and sustainable source of electricity.

Modeling of PV cell:

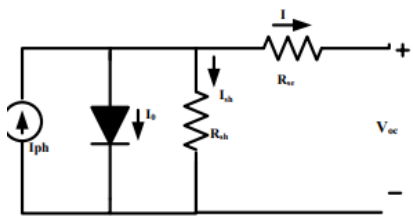


Fig. Solar cell equivalent model

PV cell modeling involves developing mathematical equations and models that describe the behavior of the cell under different operating conditions, such as changes in temperature, irradiance, and load. These models are used to simulate the electrical performance of the cell and to optimize the design and operation of PV systems.

There are several approaches to PV cell modeling, including analytical, numerical, and empirical methods. Analytical models use mathematical equations to describe the physical and electrical properties of the cell, while numerical models use computer simulations to solve complex equations and generate detailed performance predictions. Empirical models use experimental data to develop relationships between the cell's electrical output and the input parameters, such as irradiance and temperature.

The most commonly used analytical model for PV cells is the single-diode model, which describes the cell's electrical behavior using a single diode and resistor. The model includes several parameters that describe the cell's physical and electrical properties, such as the short-circuit current, open-circuit voltage, and ideality factor.

Numerical models can provide more accurate and detailed predictions of the cell's behavior, but are more computationally intensive and require more detailed input data, such as the cell's material properties and geometry.

Empirical models are often used for PV cells that are difficult to model analytically or numerically, such as cells with non-uniform doping profiles or complex geometries. These models are typically developed using experimental data and statistical analysis techniques.

Overall, PV cell modeling is an important tool for understanding and optimizing the performance of PV cells and systems, and is essential for the continued development and improvement of solar energy technology.

SOLAR MPPT TOPOLOGIES

Solar PV systems utilize a technique called maximum power point tracking (MPPT) to increase the energy production of PV modules by adjusting the system's operating point to follow the MPP of the modules. There are various MPPT topologies that can be utilized in PV systems, including:

P&O (Perturbs&Observe) is a well-known MPPT technique that intermittently perturbs the operating point of a PV system and monitors the corresponding power output. If the power output increases, the system continues in that direction, otherwise, it reverses its direction until it locates the MPP.

Incremental Conductance - This method is similar to P&O, but it also considers the change in conductance in addition to the change in power output. This approach can result in faster tracking and more accurate MPP detection.

The FOCV(Fractional open circuit voltage) is an MPPT method that estimates the Maximum Power Point (MPP) of a PV module by using its open-circuit voltage (Voc), which varies linearly with irradiance. The system adjusts the operating point to maintain a fixed fraction of the Voc, allowing it to track the MPP.

The Fractional Short-Circuit Current (FSCC) method is comparable to FOCV, but instead of open-circuit voltage (Voc), it utilizes the short-circuit current (Isc) of the PV module to estimate the MPP. It does so by adjusting the operating point to keep a constant fraction of the Isc, which varies linearly with irradiance. This enables the system to track the MPP.

Model Predictive Control (MPC) - This is a more advanced MPPT method that uses a mathematical model of the PV system to predict the future behavior of the system and optimize the operating point based on this prediction. MPC can provide faster and more accurate tracking, but requires more computational resources than other MPPT methods.

Overall, the choice of MPPT topology depends on the specific requirements and constraints of the PV system, such as the size and type of the PV modules, the operating conditions, and the desired level of performance and efficiency.

CHAPTER-IV

WIND POWER GENERATION

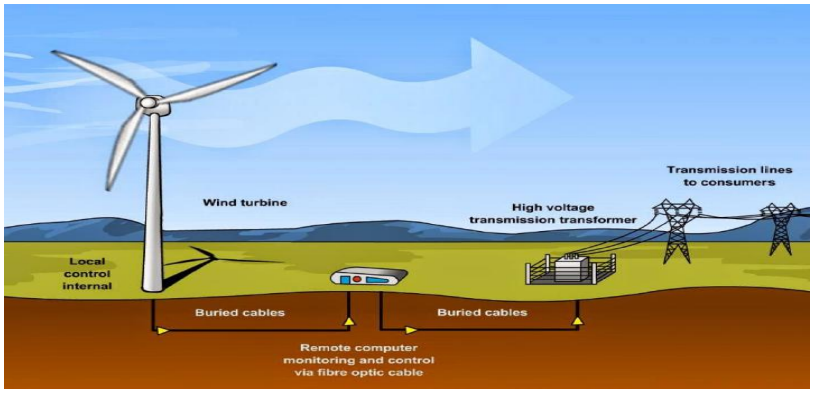


Fig. wind power generation

A wind power generation system is a renewable energy technology that transforms wind energy into electrical energy using wind turbines. Large blades that revolve when the wind blows compensate a wind turbine, which drives a generator to generate power. Wind turbine, tower, rotor blades, nacelle and control systems are the primary elements of a wind power producing system. To transform the electricity into a usable form and transport it to the grid, the system also involves electrical infrastructure like transformers and inverters.

Were there is various kinds of wind turbines, such as horizontal and vertical axis turbines, each with distinct benefits and drawbacks. Wind turbines can be installed onshore or offshore, and wind farms can range from small-scale systems for individual homes to large utility-scale projects that can power entire communities. Wind power generation systems are

considered a clean and sustainable source of energy as they do not produce greenhouse gas emissions or other pollutants associated with conventional power. However, the construction and operation of wind turbines can have impacts on wildlife, local communities, and the environment, and careful planning and management are needed to mitigate these impacts.

WIND POWER:

The production of electrical energy from the wind is referred to as wind power. Wind turbines can be used to generate. It is a sustainable and renewable source of energy. Wind power has become an increasingly popular source of renewable energy in recent years, as it is environmentally friendly and does have gas emissions. Wind turbines are typically installed in areas with consistent wind patterns, such as coastal regions or open fields. The turbines are designed with large blades that are rotated by the wind, and the rotation of the blades drives a generator that produces electricity. The electricity can be used to power homes, businesses, and entire communities, or it can be fed into the electrical grid for distribution to other locations.

There are a number of advantages to using wind power, including its sustainability, environmental friendliness, and low cost of operation. However, wind power can also face challenges, such as the need for large land areas, potential noise pollution, and the variability of wind patterns. Despite these challenges, wind power is plays an important role in the global energy work to transition to more sustainable energy sources.

Characteristics of wind power:

Here are some characteristics of wind power:

Renewable: Wind power is a renewable energy source, which means that it can be replenished naturally over time. As long as wind exists, wind power can be generated, unlike fossil fuels that are finite resources.

Abundant: Wind is a widely available resource, and wind power can be harnessed in many locations across the globe, including onshore and offshore.

Variable: Wind power output can vary greatly varying the wind speed and direction, which can impact its reliability as a primary energy source. Wind turbines also require minimum wind speeds to produce electricity, and when winds exceed a certain threshold, they can cause turbine blades to shut down for safety reasons.

Scalable: Wind power technology can be scaled to accommodate small or large energy demands, from small residential systems to large-scale wind farms.

Environmentally friendly: Wind power does not produce greenhouse gases, air pollution, or other harmful byproducts associated with fossil fuels. This makes it a cleaner alternative to traditional power sources.

Cost-effective: As technology improves, the cost of wind power continues to decline, making it increasingly competitive with other energy sources.

Land-intensive: Wind turbines require significant amounts of land, especially for larger wind farms. This can make it challenging to site wind turbines in some locations, such as densely populated areas or areas with limited land availability.

Noise pollution: Wind turbines can generate noise pollution, which can be disruptive to nearby communities. Modern turbines are designed to minimize noise, but it remains a consideration when planning wind power projects.

Wildlife impact: Wind turbines can pose a hazard to birds and bats, particularly when they are sited along migration routes. Proper siting and design can help to mitigate these impacts.

WIND GENERATOR

Fig. wind generator

A wind generator, also known as a wind turbine, is a device that converts the kinetic energy of the wind into electrical energy. Wind generators typically consist of a tower, a rotor, a generator, and other supporting components. The rotor, which is usually composed of two or three blades, is mounted on top of the tower and is designed to turn when the wind blows. As the rotor turns, it drives a generator that produces electricity. The electricity produced by the generator is typically either used immediately or stored in batteries for later use.

Wind generators come in a range of sizes, from small turbines that can power a single home to large turbines that can generate electricity for entire communities. They are often used in conjunction with other renewable energy sources, such as solar power, to provide a more reliable and consistent supply of electricity. One of the advantages of wind generators is that they are a clean and renewable source of energy, as they do not produce greenhouse gases or

other pollutants that can harm the environment. However, they can also have negative impacts on wildlife and their installation and maintenance can be costly.

Types of Wind Turbine Generator

There are several types of wind generators, each with its own unique design and application. Here are some of the most common types:

1. **Single-phase AC generators**: This type of wind generator produces alternating current (AC) power with a single-phase output. These generators are typically used in small-scale wind power systems, such as those used to power homes or small businesses.
2. **Three-phase AC generators**: Three-phase AC generators produce AC power with a three-phase output, which is more efficient and provides a smoother power supply than single-phase generators. These generators are typically used in larger wind power systems, such as those used to power communities or industrial facilities.
3. **Direct-drive generators**: Direct-drive generators eliminate the need for a gearbox by connecting the rotor directly to the generator. This design simplifies the system, reduces maintenance, and increases efficiency. Direct-drive generators are commonly used in larger wind turbines.
4. **Gear-driven generators**: Gear-driven generators use a gearbox to connect the rotor to the generator. This allows the generator to spin faster than the rotor and increases the efficiency of the system. Gear-driven generators are commonly used in smaller wind turbines.
5. **Permanent magnet generators**: Permanent magnet generators use powerful magnets to generate a magnetic field that produces electrical power. These generators are typically more efficient and require less maintenance than other types of generators.
6. **Induction generators**: Induction generators use electromagnetic induction to generate electrical power. They are commonly used in larger wind turbines and are often less expensive than other types of generators.

Asynchronous generator

An asynchronous wind generator, also known as an induction generator, is a type of generator used in wind turbines to convert the kinetic energy of wind into electrical energy. Asynchronous generators are often used in larger wind turbines because they are more cost-effective than other types of generators, such as synchronous generators. Asynchronous generators work by using electromagnetic induction to generate electrical power. They consist of a rotor, which is mounted on a shaft, and a stator, which is a stationary component that contains the windings of the generator. When the rotor is turned by the force of the wind, it

creates a rotating magnetic field that induces a current in the stator windings, which generates electrical power.

One of the key advantages of asynchronous generators is their simplicity. They do not require a direct connection to the grid and can operate independently, which makes them useful in remote locations. They also do not require the use of brushes or slip rings, which reduces maintenance requirements. However, asynchronous generators do have some drawbacks. They are less efficient than synchronous generators at low wind speeds, which can limit their performance in some wind conditions. They also require a capacitor bank to provide the necessary reactive power to operate, which can add to the cost of the system.

DESIGN OF WIND TURBINE:

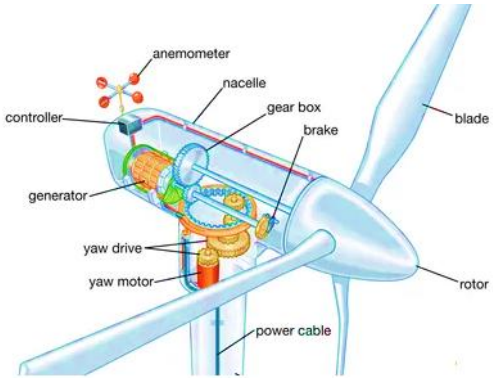


Fig. wind turbine

The design of a wind turbine involves several components that work together to capture the kinetic energy of the wind and convert it into electrical energy. Here are the main parts of a wind turbine:

The design of a wind turbine

Rotor blades: The blades are the most visible part of a wind turbine. They are designed to capture the wind and convert its kinetic energy into rotational energy. The blades are typically made of lightweight and durable materials, such as fiberglass or carbon fiber.

Rotor hub: The rotor hub is the central component that connects the rotor blades to the main shaft. The hub allows the blades to rotate and change pitch to optimize their angle of attack to the wind.

Nacelle: The nacelle is the housing that places at the top of the tower and contains the gearbox, generator, and other components that convert the rotational energy of the rotor into electrical energy. The nacelle also includes the yaw system, which enables the turbine to turn and face the wind.

Tower: The tower is the support structure that holds the rotor and nacelle at a height that maximizes the wind resource. Towers can be constructed from steel, concrete, or other materials.

Control system: The control system is the brain of the wind turbine. It monitors the wind direction and speed, adjusts the rotor blades to optimize their angle of attack, and controls the power output of the turbine.

The structure of a wind turbine is optimized for specific wind conditions, such as wind speed and turbulence. Modern wind turbines are designed to be highly efficient, with rotors that can adjust to varying wind conditions and control systems that can optimize power output. The size of wind turbines has also increased significantly in recent years, with some models reaching heights of over 200 meters and rotor diameters of up to 220 meters.

Wind power generation control:

Wind power generation control involves managing the output of wind turbines to maintain a stable and reliable supply of electrical energy. Here are some of the key aspects of wind power generation control:

Turbine control: Wind turbines are equipped with sensors and control systems that monitor wind speed and direction, and adjust the pitch and orientation of the blades to maximize energy capture. The control system also monitors the performance of the turbine and can shut it down if there are safety or performance issues.

Power output control: The power output of wind turbines can be controlled by adjusting the pitch angle of the blades or by limiting the amount of power that is fed into the electrical grid. This is important for maintaining grid stability and avoiding power imbalances that can cause blackouts or damage to the grid.

Grid integration: Wind turbines are typically connected to the electrical grid through power electronics that convert the variable output of the turbine into a stable, high-quality electrical signal that can be fed into the grid. Grid integration also involves managing the interaction between wind turbines and other power sources, such as conventional power plants or other renewable energy sources.

Forecasting and scheduling: Wind power generation control also involves forecasting the expected output of wind turbines based on weather conditions and scheduling their operation to meet the needs of the grid. This helps to avoid overloading the grid and ensures that wind power is available when it is needed.

Energy storage: Energy storage technologies, such as batteries or pumped hydro storage, can be used to store excess wind power and release it during times of high demand. This helps to improve the reliability and stability of wind power generation and makes it easier to integrate into the electrical grid.

Overall, wind power generation control is a complex process that involves balancing the variable output of wind turbines with the needs of the electrical grid. As wind power continues to play an increasingly important role in the global energy mix, the development of new control technologies and strategies will be critical for ensuring its reliability and effectiveness.

CHAPTER-V

Buck converter

History

The buck converter was first introduced in the 1960s as a solution to the problem of inefficient power supplies in early electronic devices. Prior to the development of the buck converter, linear regulators were commonly used to regulate voltage, but they suffered from poor efficiency and generated significant heat.

The first practical implementation of the buck converter was the "synchronous buck regulator" developed by engineers at Bell Labs in 1966. This regulator used a synchronous switching transistor to control the voltage, and it achieved higher efficiency and improved voltage regulation compared to previous solutions.

In the following decades, the buck converter continued to evolve and improve, with advancements in control circuits, switching transistors, and power semiconductors. In the 1980s, the introduction of pulse-width modulation (PWM) control circuits helped to improve the accuracy and efficiency of buck converters.

Today, buck converters are widely used in a variety of electronic devices, from smartphones and laptops to automotive and industrial applications. They are known for their high efficiency, small size, and ability to provide stable voltage regulation, making them a critical component in modern electronics.

A buck converter is a type of DC-to-DC power converter that is commonly used to step down a DC voltage from a higher level to a lower level. The term "buck" refers to the fact that the output voltage is reduced, or "bucked down", from the input voltage.

Introduction

The buck converter consists of a switching transistor, an inductor, a diode, and a capacitor. During operation, the switching transistor turns on and off at a high frequency, and the inductor and capacitor work together to store and release energy in response to changes in current. The output voltage is regulated by adjusting the duty cycle of the switching transistor, which controls the amount of time it spends in the "on" state versus the "off" state.

Buck converters are commonly used in power supplies for electronic devices, such as laptops, smartphones, and other portable devices, as well as in automotive and industrial applications. They are known for their high efficiency and ability to provide a stable output voltage even when the input voltage varies.

Overall, the buck converter is a versatile and important component in modern electronics and plays a key role in the efficient delivery of power to a wide range of devices and applications.

What is the need of buck converter?

There are several reasons why a buck converter may be needed in a circuit:

Voltage regulation: A buck converter can be used to regulate the output voltage of a power supply. By controlling the duty cycle of the switching transistor, the output voltage can be maintained at a constant level, even if the input voltage varies.

Energy efficiency: Buck converters are highly efficient at converting DC voltage from a higher level to a lower level. This is because the inductor and capacitor used in the circuit store and release energy, which helps to reduce power loss and heat generation.

Space saving: Buck converters are typically smaller and lighter than linear regulators, which makes them well-suited for use in portable devices and other applications where space is limited.

Battery life extension: Buck converters are commonly used in battery-powered devices to extend battery life. By converting the voltage to a lower level, less energy is wasted as heat, which means the battery can last longer.

Voltage matching: In some cases, the load may require a lower voltage than the available input voltage. A buck converter can be used to match the input voltage to the required output voltage for the load.

Overall, the use of a buck converter can help to improve the efficiency, reliability, and performance of a power supply circuit.

What is buck converter?

A buck converter is a type of DC-to-DC power converter that steps down a DC voltage from a higher level to a lower level. The output voltage of a buck converter is lower than the input voltage, and the conversion is achieved by controlling the duty cycle of a switching transistor that is connected to an inductor and a capacitor.

During operation, the switching transistor turns on and off at a high frequency, and the inductor stores and releases energy in response to the changes in current. The output voltage is regulated by adjusting the duty cycle of the transistor, which controls the amount of time it spends in the "on" state versus the "off" state.

Buck converters are commonly used in power supplies for electronic devices, such as laptops, smartphones, and other portable devices, as well as in automotive and industrial applications. They are known for their high efficiency and ability to provide a stable output voltage even when the input voltage varies.

Working

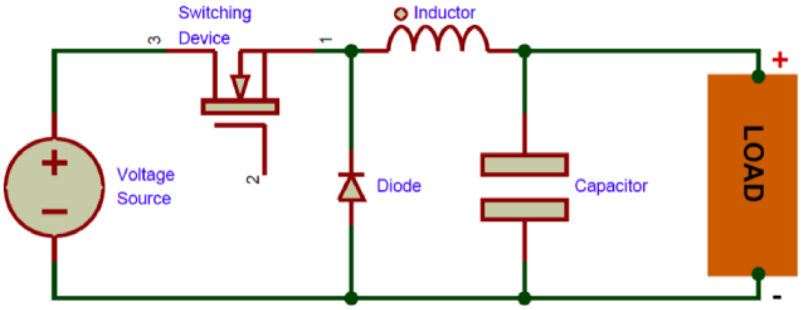


Fig. buck converter

The working of a buck converter can be explained in the following steps:

During the first phase, the switching transistor in the buck converter is turned on, allowing current to flow from the input voltage source (usually a battery or power supply) through the inductor and to the output capacitor and load.

As the current flows through the inductor, it stores energy in the form of a magnetic field. The energy stored in the inductor increases with time as long as the switching transistor remains on.

In the second phase, the switching transistor is turned off, causing the magnetic field in the inductor to collapse. As the magnetic field collapses, the energy stored in the inductor is released and current continues to flow through the circuit, but now through the output capacitor and the load.

The output capacitor and load absorb the energy released by the inductor, which helps to regulate the output voltage and provide a steady supply of power to the load.

The switching transistor then turns back on, starting the first phase of the cycle again. The duty cycle, or the percentage of time that the switching transistor is on, determines the output voltage of the buck converter. By adjusting the duty cycle, the output voltage can be regulated to the desired level.

Overall, the buck converter converts a higher input voltage to a lower output voltage while maintaining a stable supply of power to the load. The efficiency of the buck converter is due to the inductor and capacitor, which store and release energy, reducing power loss and heat generation.

Advantages

There are several advantages of using a buck converter in a circuit:

High efficiency: Buck converters are highly efficient at converting DC voltage from a higher level to a lower level. The use of an inductor and capacitor in the circuit allows for energy storage and release, which helps to reduce power loss and heat generation.

Small size: Buck converters are typically smaller and lighter than linear regulators, making them well-suited for use in portable devices and other applications where space is limited.

Voltage regulation: Buck converters can regulate the output voltage of a power supply, maintaining a constant output voltage even if the input voltage varies.

Improved battery life: Buck converters are commonly used in battery-powered devices to extend battery life. By converting the voltage to a lower level, less energy is wasted as heat, which means the battery can last longer.

Cost-effective: Buck converters are relatively inexpensive and easy to manufacture, making them a cost-effective solution for power supply circuits.

High current capability: Buck converters are capable of handling high current loads, making them suitable for a wide range of applications.

Overall, the use of a buck converter can help to improve the efficiency, reliability, and performance of a power supply circuit, making it a popular choice for electronic devices, automotive systems, and other applications.

Drawbacks

While there are many advantages to using a buck converter in a circuit, there are also some potential drawbacks that should be considered:

Limited voltage reduction: Buck converters can only reduce the input voltage to a certain extent, typically up to around 50% of the input voltage. If a greater voltage reduction is required, a different type of DC-DC converter may be needed.

Output voltage ripple: The switching action of the buck converter can cause voltage ripple on the output voltage, which can be problematic for sensitive circuits or analog applications. However, this can be mitigated through proper filtering and design techniques.

EMI/RFI noise: The fast switching action of the buck converter can also generate electromagnetic interference (EMI) or radio frequency interference (RFI), which can affect other circuits or devices in the vicinity. Again, this can be addressed through proper design and filtering techniques.

Complexity: Buck converters can be more complex than linear regulators or other voltage regulators, requiring more components and more precise design considerations. This can make them more difficult and costly to implement in some applications.

Efficiency vs. load current: The efficiency of the buck converter can decrease at low load currents, which can result in lower overall efficiency and wasted power. Careful design and component selection can help mitigate this issue.

Overall, the disadvantages of the buck converter are generally outweighed by its many advantages, making it a popular and widely used solution for voltage regulation in a variety of applications.

Applications

Buck converters have a wide range of applications in various industries, including:

Portable devices: Buck converters are commonly used in portable electronic devices such as smartphones, laptops, and tablets. They help to extend battery life by converting the input voltage to a lower level, which reduces power loss and heat generation.

Automotive industry: Buck converters are used in automotive systems to regulate voltage and provide stable power to electronic components such as headlights, sensors, and entertainment systems.

Industrial equipment: Buck converters are used in industrial applications such as motor control, robotics, and automation systems. They provide efficient and reliable power conversion, helping to improve the performance and longevity of these systems.

Renewable energy systems: Buck converters are used in renewable energy systems such as solar panels and wind turbines to convert DC voltage from the source to a usable AC voltage. They also help to regulate voltage and improve system efficiency.

LED lighting: Buck converters are used in LED lighting systems to regulate voltage and provide stable power to the LEDs. They also help to improve energy efficiency and reduce heat generation.

Overall, the buck converter is a versatile and widely used solution for voltage regulation and power conversion in a variety of applications, from consumer electronics to industrial equipment and renewable energy systems.

BOOST CONVERETER

HISTORY

The Boost Converter, also known as the Step-Up Converter, is a type of DC-DC converter that has been in use since the early 1920s. It was first used as a voltage regulator for vacuum-tube amplifiers.

The first Boost Converter was a self-oscillating type, which was introduced by Henry Chireix in 1928. Chireix was a French engineer who worked for the French Broadcasting Company. He used the Boost Converter to regulate the voltage of the radio transmitter tubes.

In the 1960s, the Boost Converter was further developed and improved by Robert Mammano and Richard Baker. They introduced the PWM (Pulse-Width Modulation) technique for controlling the output voltage of the Boost Converter. This made the Boost Converter more efficient and allowed it to be used in a wider range of applications.

The Boost Converter became more popular in the 1970s and 1980s with the development of semiconductor devices such as the MOSFET (Metal-Oxide-Semiconductor Field-Effect

Transistor) and the IGBT (Insulated Gate Bipolar Transistor). These devices made it possible to build more efficient and reliable Boost Converters.

Today, Boost Converters are widely used in a variety of applications, such as power supplies for electronic devices, battery chargers, LED drivers, and renewable energy systems. They have become an essential part of modern power electronics and continue to be improved and developed.

WHAT IS BOOST CONVERTER

A Boost Converter is a type of DC-DC converter that is used to step up the input voltage to a higher output voltage level. It is also known as a Step-Up Converter.

The Boost Converter works by storing energy in an inductor during the ON period of a switching transistor and then releasing this stored energy to the output during the OFF period of the transistor. This causes the voltage across the inductor to increase, which results in a higher output voltage level than the input voltage.

The Boost Converter consists of an input capacitor, an inductor, a switching transistor, a diode, and an output capacitor. The switching transistor is typically controlled by a PWM (Pulse-Width Modulation) signal, which regulates the amount of energy transferred from the input to the output.

The Boost Converter is commonly used in applications where the input voltage is too low to power a load, such as in battery-powered devices, solar panels, and other renewable energy sources. It is also used in power supplies for electronic devices, where a higher voltage level is required for efficient operation.

One of the advantages of the Boost Converter is its simplicity and efficiency, as it requires fewer components and can achieve high conversion efficiency.

WHAT IS THE NEED OF BOOST CONVERTER

A Boost Converter is needed when a load requires a higher voltage level than the input voltage can provide. In such cases, a Boost Converter can step up the input voltage to a higher output voltage level, which can then power the load.

For example, in battery-powered devices, the battery voltage may not be sufficient to power the load. A Boost Converter can be used to step up the battery voltage to a higher voltage level, which can then power the load efficiently.

Similarly, in renewable energy systems such as solar panels, the output voltage of the panel may vary depending on the amount of sunlight. A Boost Converter can be used to step up the voltage of the panel to a higher level, which can then be used to power the load or charge a battery.

In power supplies for electronic devices, a Boost Converter can be used to provide a stable output voltage regardless of the input voltage level. This is important because electronic devices require a stable and regulated power supply to operate efficiently and reliably.

Overall, the need for a Boost Converter arises whenever a load requires a higher voltage level than the input voltage can provide. It is an essential component in modern power electronics and is widely used in a variety of applications.

WORKING

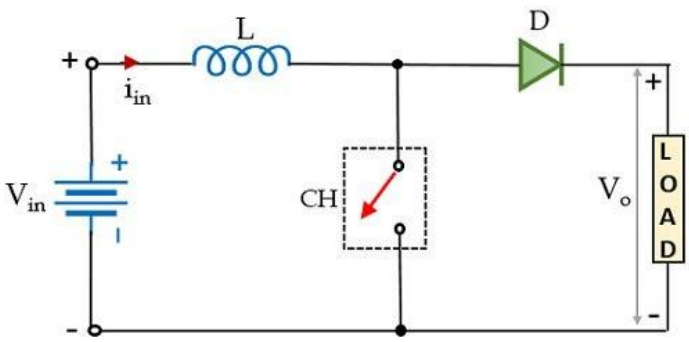


Fig. boost converter

The Boost Converter is a type of DC-DC converter that can step up the input voltage to a higher output voltage level. The working principle of a Boost Converter can be explained in the following steps:

During the ON period of the switching transistor, the input voltage is applied to the inductor, and current starts flowing through the inductor. The current in the inductor increases linearly, storing energy in the inductor.

When the switching transistor turns OFF, the current in the inductor cannot change instantaneously. The energy stored in the inductor is then transferred to the output capacitor through the diode, resulting in an increased output voltage.

The inductor current then starts decreasing, and the energy stored in the inductor is again used to charge the output capacitor during the next ON period of the switching transistor.

The output voltage is regulated by controlling the duty cycle of the switching transistor using a PWM signal. By adjusting the duty cycle, the amount of energy transferred from the input to the output can be controlled, resulting in a regulated output voltage.

Overall, the Boost Converter works by storing energy in the inductor during the ON period of the switching transistor and then releasing this energy to the output during the OFF period of the transistor. This results in a higher output voltage level than the input voltage. The Boost Converter is a simple and efficient way to step up the input voltage and is widely used in a variety of applications.

ADVANTAGES

Boost Converter, also known as Step-Up Converter, offers several advantages in comparison to other DC-DC converters:

High Efficiency: Boost Converter has high efficiency due to the use of switching techniques to regulate the output voltage. It can achieve efficiencies of up to 95%, which makes it highly desirable for low-power applications.

Simple Circuitry: Boost Converter circuitry is simple and consists of only a few components, including an inductor, a diode, a capacitor, and a switching transistor. This makes it easy to design and implement.

Low Cost: Due to the simple circuitry and fewer components, Boost Converters are relatively low-cost compared to other DC-DC converters.

Wide Range of Input Voltages: Boost Converters can operate over a wide range of input voltages, making them suitable for applications with varying input voltage levels, such as battery-powered devices and renewable energy systems.

Compact Size: Boost Converters are relatively compact due to their simple circuitry and fewer components. This makes them ideal for space-constrained applications.

Regulated Output Voltage: Boost Converters can provide a regulated output voltage, making them suitable for applications that require a stable and regulated power supply.

Overall, Boost Converters offer several advantages, including high efficiency, low cost, simple circuitry, wide input voltage range, compact size, and regulated output voltage. These advantages make Boost Converters highly desirable for a wide range of applications in modern power electronics.

DISADVANTAGES

Although Boost Converters offer several advantages, they also have some disadvantages that should be considered:

Output Voltage Ripple: Boost Converters can have a high output voltage ripple due to the switching action of the transistor. This can cause unwanted noise in the circuit and affect the performance of the load.

Limited Load Current: Boost Converters are typically designed for low-power applications and have a limited load current capability. If the load current exceeds the converter's rating, it can lead to overloading and damage to the converter.

Complex Control: Boost Converters require complex control circuitry to regulate the output voltage and maintain the desired performance. This can make them challenging to design and implement.

EMI and RFI Noise: Boost Converters can generate electromagnetic interference (EMI) and radio-frequency interference (RFI) noise due to the switching action of the transistor. This can affect the performance of other nearby circuits and require additional filtering and shielding.

Inrush Current: During start up, Boost Converters can experience high inrush current, which can damage the input capacitor or the converter itself if it is not designed to handle such current.

Overall, while Boost Converters offer several advantages, they also have some disadvantages that must be considered in the design and implementation of power electronics applications. Proper selection of components, careful design, and good system integration can help mitigate these disadvantages and ensure optimal performance.

APPLICATIONS

Boost Converters find applications in a variety of power electronics applications where a higher output voltage is required from a lower input voltage. Some of the common applications of Boost Converters include:

Battery-Powered Devices: Boost Converters are commonly used in battery-powered devices such as portable electronics, mobile phones, and laptops to step up the battery voltage to a higher voltage required by the device.

LED Lighting: Boost Converters are used in LED lighting applications to provide a constant current to the LED and step up the voltage to match the LED requirements.

Renewable Energy Systems: Boost Converters are used in renewable energy systems such as solar and wind power systems to step up the voltage from the low voltage generated by the renewable source to a higher voltage required by the load or the grid.

Automotive Systems: Boost Converters are used in automotive systems to step up the battery voltage to a higher voltage required by the vehicle's electronics and accessories.

Medical Equipment: Boost Converters are used in medical equipment such as ultrasound machines, X-ray machines, and MRI machines to step up the voltage and provide a stable and regulated power supply.

Aerospace and Military Applications: Boost Converters are used in aerospace and military applications such as avionics, guidance systems, and radar systems to provide a stable and regulated power supply from a variable input voltage.

Overall, Boost Converters find applications in a wide range of power electronics applications where a higher output voltage is required from a lower input voltage.

CHAPTER-VI

Maximum Power Point Tracking

History

Maximum Power Point Tracking (MPPT) is a technique used in photovoltaic (PV) systems to extract maximum power from a solar panel. The idea of using MPPT in solar power systems has been around for several decades. However, its implementation has evolved over time with the development of technology.

In the early days of solar power systems, there were no MPPT controllers. The solar panels were directly connected to the battery bank, which resulted in a significant power loss. As the load increased, the voltage of the solar panel dropped, leading to a reduced power output. This problem led to the development of the first-generation MPPT controllers.

The first-generation MPPT controllers were based on analog circuits. They used a variable resistor to vary the voltage of the solar panel and track the maximum power point. These controllers were not very efficient, and their accuracy was limited.

With the advancement of digital technology, the second-generation MPPT controllers were developed. These controllers used microprocessors to control the voltage of the solar panel and extract maximum power. They were more efficient than the first-generation controllers and had better accuracy.

In the late 1990s, the third-generation MPPT controllers were developed. These controllers used a technique called pulse width modulation (PWM) to control the voltage of the solar panel. PWM allowed the controller to rapidly switch the voltage on and off, resulting in a more efficient power output.

In recent years, the fourth-generation MPPT controllers have been developed, which use advanced algorithms and machine learning techniques to optimize the power output of solar panels. These controllers can adjust their parameters based on environmental conditions, such as temperature and irradiance, to maximize power output.

Today, MPPT controllers are widely used in solar power systems, from small residential systems to large commercial systems. They have become an essential component in maximizing the efficiency and effectiveness of solar power systems.

What is solar mppt

Maximum Power Point Tracking (MPPT) is a technique used in photovoltaic (PV) systems to extract maximum power from a solar panel. The amount of power generated by a solar panel depends on various factors such as the intensity of the sunlight, the temperature, and the load on the panel. The maximum power point (MPP) of a solar panel is the point at which the panel can produce the maximum power output under specific conditions.

An MPPT controller is an electronic device that is used in solar power systems to track the MPP of a solar panel and optimize its power output. The MPPT controller adjusts the voltage and current supplied to the load in such a way that it extracts the maximum power from the panel. This is achieved by continuously monitoring the voltage and current of the panel and adjusting the load to ensure that the panel operates at its MPP.

MPPT controllers work by using a DC-to-DC converter that adjusts the voltage and current of the panel to match the load. The controller monitors the voltage and current of the panel and uses a feedback loop to adjust the output voltage of the converter until it reaches the MPP. The controller continuously tracks the MPP of the panel, adjusting the output voltage of the converter to ensure that the panel operates at its maximum efficiency.

MPPT controllers are used in a wide range of solar power systems, from small residential systems to large commercial systems. They have become an essential component in maximizing the efficiency and effectiveness of solar power systems.

Advantages of solar mppt

The use of Maximum Power Point Tracking (MPPT) controllers in solar power systems has several advantages:

Higher efficiency: MPPT controllers can extract the maximum power from the solar panel, resulting in higher efficiency and increased power output. This means that more power can be generated from the same amount of sunlight, reducing the overall cost of the system.

Better performance in low light conditions: MPPT controllers can adjust the voltage and current of the panel to match the load, allowing the panel to operate at its maximum efficiency even in low light conditions. This means that more power can be generated even on cloudy days.

Longer lifespan of the battery: MPPT controllers prevent overcharging and over-discharging of the battery, which can extend its lifespan. The controller adjusts the charging rate to ensure that the battery is charged to its optimal level and maintains a constant voltage to prevent over-discharging.

Flexibility: MPPT controllers can be used with different types of solar panels, including those with varying voltage and current ratings. This makes them more flexible and adaptable to different types of solar power systems.

Remote monitoring and control: MPPT controllers can be remotely monitored and controlled using a computer or smartphone. This allows the user to monitor the performance of the solar power system and adjust the settings as needed.

Overall, the use of MPPT controllers in solar power systems can result in higher efficiency, increased power output, longer lifespan of the battery, and better performance in low light conditions. This makes them an essential component in maximizing the effectiveness and efficiency of solar power systems.

Applications of solar mppt

Maximum Power Point Tracking (MPPT) controllers are used in a wide range of solar power applications, including:

Residential solar systems: MPPT controllers can be used in small residential solar systems to optimize power output and increase efficiency. This can help reduce electricity bills and provide a sustainable source of energy for the home.

Commercial solar systems: MPPT controllers can be used in large commercial solar systems to maximize power output and improve energy efficiency. This can help reduce operating costs and provide a sustainable source of energy for businesses.

Solar-powered street lighting: MPPT controllers can be used in solar-powered street lighting systems to optimize power output and increase efficiency. This can help reduce electricity bills and provide a sustainable and cost-effective lighting solution for streets and public spaces.

Solar-powered water pumping: MPPT controllers can be used in solar-powered water pumping systems to maximize power output and increase efficiency. This can help reduce operating costs and provide a sustainable source of water for agricultural and rural areas.

Remote monitoring systems: MPPT controllers can be used in remote monitoring systems that use solar power to provide electricity to monitoring equipment. This can help reduce the need for traditional power sources and provide a sustainable solution for monitoring and surveillance in remote areas.

Overall, the applications of MPPT controllers in solar power systems are diverse and can provide sustainable and cost-effective solutions for a wide range of industries and applications.

Types for mppt

There are different types of MPPT (Maximum Power Point Tracking) algorithms used in solar photovoltaic systems. Here are some of the most common types:

Perturb and Observe (P&O): This is the most widely used MPPT algorithm. It works by continuously perturbing the operating point of the solar panel and observing the effect on the output power. The algorithm then determines the direction in which the power changes and adjusts the operating point in that direction until the maximum power point is reached.

Incremental Conductance: This algorithm is similar to P&O, but it also takes into account the change in the slope of the power-voltage curve. It adjusts the operating point to follow the slope of the curve, which helps to track the maximum power point more accurately.

Fractional Open Circuit Voltage: This algorithm is based on the open-circuit voltage of the solar panel, which is proportional to the amount of sunlight hitting the panel. It calculates the fractional open-circuit voltage and uses it to estimate the maximum power point.

Artificial Neural Network (ANN): This algorithm uses a neural network to learn the relationship between the input variables (such as irradiance and temperature) and the output power. Once the network is trained, it can predict the maximum power point based on the input variables.

Fuzzy Logic Control: This algorithm uses fuzzy logic to adjust the operating point of the solar panel. It takes into account the change in the slope of the power-voltage curve and the rate of change of the irradiance to track the maximum power point.

The choice of MPPT algorithm depends on various factors such as the type of solar panel, the complexity of the system, and the desired performance.

CHAPTER-VII

Artificial Neural Network (ANN) controllers

History

Artificial Neural Network (ANN) controllers have a long history, dating back to the early days of computer science and artificial intelligence. The first ANN controller was developed in the 1940s by Warren McCulloch and Walter Pitts, who created a simple model of the brain using interconnected artificial neurons.

Over the years, ANNs have evolved and become more complex, with the development of new algorithms, architectures, and training techniques. In the 1960s and 1970s, researchers developed the backpropagation algorithm, which is still widely used today for training ANNs. In the 1980s and 1990s, researchers developed new architectures, such as the multilayer perceptron and recurrent neural networks, which allowed ANNs to handle more complex problems.

ANN controllers have been used in a wide range of applications, including control systems, robotics, image processing, and natural language processing. In the 1990s, researchers began to explore the use of ANNs in adaptive control systems, which could learn and adapt to

changing environments. This led to the development of techniques such as reinforcement learning, which uses ANNs to learn optimal control policies through trial and error.

In recent years, ANNs have become even more powerful, thanks to advances in computing power, data storage, and machine learning techniques. Today, ANNs are used in a wide range of industries and applications, from self-driving cars and industrial automation to finance and healthcare. With ongoing research and development, the capabilities of ANN controllers are likely to continue to expand, making them even more valuable tools for solving complex problems.

What is ANN CONTROLLER?

An Artificial Neural Network (ANN) controller is a type of control system that uses an ANN to control a process or system. The ANN controller is a computational model that is designed to simulate the behavior of the human brain and consists of interconnected artificial neurons that can learn and adapt to changing conditions.

The ANN controller receives input signals from sensors that measure various parameters of the system being controlled, such as temperature, pressure, or speed. These inputs are then processed by the ANN, which generates an output signal that is sent to an actuator or other control device to adjust the system's behavior.

ANN controllers can be used in a wide range of applications, including industrial automation, robotics, and self-driving vehicles. They are particularly useful in situations where the system being controlled is complex and difficult to model using traditional control techniques.

One of the advantages of ANN controllers is their ability to learn and adapt to changing conditions. During the training phase, the ANN is presented with input-output pairs and adjusts its weights and biases to minimize the error between the actual and predicted outputs. This process allows the ANN to learn the optimal control policy for the system being controlled, which can then be used to adjust the system's behavior in real-time.

Overall, ANN controllers offer a powerful and flexible approach to control that can be used in a wide range of applications. With ongoing research and development, ANN controllers are likely to become even more powerful tools for solving complex control problems.

What is the need of ANN CONTROLLER?

The need for an Artificial Neural Network (ANN) controller arises in situations where traditional control techniques are not effective due to the complexity of the system being controlled or the unpredictability of the operating environment. In such cases, an ANN controller can offer several advantages:

Ability to learn and adapt: ANNs are capable of learning from data and adapting to changing conditions, making them well-suited to systems that are dynamic or unpredictable. This means that an ANN controller can adjust its behavior in real-time, based on feedback from sensors, to optimize performance and minimize errors.

Nonlinear control: Many real-world systems exhibit nonlinear behavior, which means that their response is not proportional to the input signal. ANNs are well-suited to handling nonlinear systems since they can approximate any function to a desired degree of accuracy.

Fault tolerance: ANNs are capable of fault tolerance, meaning that they can continue to operate even if some of their components fail. This makes them useful for critical applications where downtime must be minimized.

Parallel processing: ANNs are capable of parallel processing, meaning that they can perform many computations simultaneously. This can lead to faster and more efficient control of complex systems.

Overall, the need for an ANN controller arises when traditional control techniques are inadequate to handle the complexity, nonlinearity, or unpredictability of a system. ANNs offer a flexible and powerful approach to control that can adapt to changing conditions and optimize system performance.

Working of ANN controller

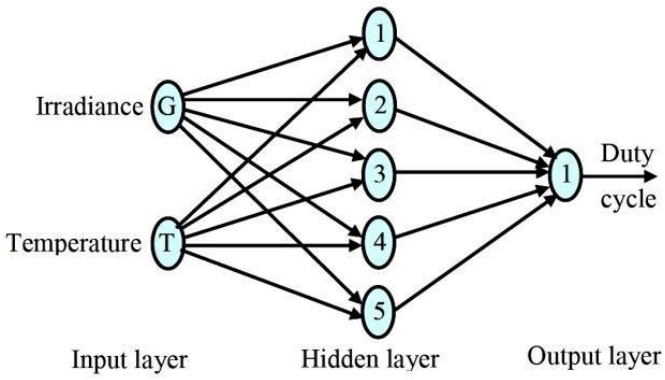


Fig. structure of ANN controller

An Artificial Neural Network (ANN) controller is a type of control system that utilizes a neural network to learn and make decisions based on input data. The following are the general steps involved in the working of an ANN controller:

Data Collection: The first step in the process is to collect data from the environment or system being controlled. The data collected will be used as input to the neural network.

Pre-processing: The collected data is preprocessed to make it suitable for input into the neural network. This may include scaling, normalization, and other data preprocessing techniques.

Neural Network Architecture Design: The neural network architecture is designed by selecting the number of layers, the number of nodes in each layer, and the activation functions to be used.

Training: The neural network is trained using a dataset that contains input-output pairs. The goal is to adjust the network's weights and biases so that the network can accurately predict the output for new inputs.

Validation: The network's performance is validated using a separate dataset that was not used for training. This step ensures that the network is not overfitting to the training data and can generalize to new data.

Deployment: The trained neural network is deployed in the control system to make decisions based on the input data. The neural network's output is used to control the system being controlled, such as a robot or a manufacturing process.

Monitoring and Optimization: The system's performance is monitored, and the neural network is optimized to improve its accuracy and efficiency continually. This step may involve adjusting the neural network's architecture, training it on more data, or fine-tuning its parameters.

Overall, the working of an ANN controller involves collecting and preprocessing data, designing and training a neural network, deploying it in the control system, and continually monitoring and optimizing its performance to improve the system's efficiency and accuracy.

Advantages of ANN controller

Artificial Neural Network (ANN) controllers offer several advantages over other control systems, including:

Non-linearity: ANN controllers can model and control non-linear systems, which are difficult or impossible to model using traditional control systems. This makes ANN controllers suitable for a wide range of applications, including robotics, manufacturing, and finance.

Adaptability: ANNs can learn from experience and adapt to changing environments, making them ideal for control systems that need to adjust to new or unpredictable situations. This adaptability also means that ANN controllers can continue to improve their performance over time.

Fault tolerance: ANN controllers are capable of functioning even if some of their components fail or are damaged. This makes them more robust and reliable than traditional control systems.

Parallel processing: ANNs can process multiple inputs simultaneously, allowing them to control complex systems quickly and efficiently.

Reduced human intervention: ANN controllers can operate autonomously, reducing the need for human intervention and supervision. This is particularly useful in applications such as robotics, where the system must be able to function independently.

Real-time processing: ANNs can process data in real-time, making them suitable for applications that require immediate response and decision-making.

Overall, the advantages of ANN controllers make them an attractive option for a wide range of control applications, from robotics and manufacturing to finance and healthcare.

Drawbacks of ANN controller

While Artificial Neural Network (ANN) controllers offer many advantages, they also have some drawbacks that should be considered:

Training requirements: ANN controllers require large amounts of data for training, and the training process can be computationally intensive and time-consuming. This can make it challenging to develop and optimize the network for a specific application.

Black-box nature: ANNs are often considered as black-box models, meaning it can be difficult to interpret the decision-making process or understand how the network arrived at a particular output. This can make it challenging to troubleshoot or debug the system when problems arise.

Overfitting: ANN controllers can be susceptible to overfitting, where the network becomes too specialized to the training data and performs poorly on new data. This can be mitigated by using appropriate regularization techniques and validation strategies.

Limited transferability: ANNs trained for one task may not be easily transferable to a new task or domain. This means that retraining or significant modification of the network may be necessary when applying it to a new application or domain.

Complexity: ANN controllers can be complex and difficult to design and optimize, especially for large-scale or complex systems. This can lead to higher development and maintenance costs.

Hardware requirements: ANN controllers may require specialized hardware to support the computational demands of training and inference. This can increase the cost and complexity of deploying the system.

Overall, ANN controllers have many benefits but also some challenges that need to be considered carefully when designing and implementing a control system.

Applications of ANN controller

Artificial Neural Network (ANN) controllers have a wide range of applications in various fields, including:

Robotics: ANN controllers can be used to control the movements and behaviors of robots. For example, ANN controllers can be used to control the movements of an autonomous vehicle or a robot arm in manufacturing.

Finance: ANN controllers can be used for financial forecasting, risk management, and fraud detection. For example, ANN controllers can be used to predict stock prices or credit risk.

Manufacturing: ANN controllers can be used to control and optimize manufacturing processes, such as predicting the quality of a product or controlling the flow of materials in a production line.

Healthcare: ANN controllers can be used for medical diagnosis, image recognition, and drug discovery. For example, ANN controllers can be used to predict the onset of a disease or to identify cancer cells in medical images.

Agriculture: ANN controllers can be used to optimize crop yields, predict weather patterns, and manage irrigation systems.

Energy: ANN controllers can be used to optimize energy consumption and production in power plants and smart grids.

Transportation: ANN controllers can be used to optimize traffic flow, control autonomous vehicles, and manage transportation systems.

Overall, ANN controllers have a wide range of applications and are used in various industries to improve efficiency, accuracy, and decision-making

CHAPTER-VIII

SYSTEM DISCREPTION

SOLAR/WIND SYSTEM CHARACTERISTICS

In order to explain and build the hybrid system presented in this paper, solar PV panels and wind turbines combine to create a highly reliable hybrid energy system that can harvest renewable energy under various weather situations [5, 6]. When the sun is shining and the wind isn't too strong, solar panels will generate energy; nevertheless, when severe weather strikes and the sky is cloudy, wind turbines become crucial. To maximize the utilization of renewable energy resources, the suggested solar/wind hybrid power systems combine both sources of electricity with a bank of batteries to store the energy. In the case of extra energy, by using an inverter, the extra energy from the solar part can feed the power grid [7, 8]. In addition, to design a power plant, a machine learning model is required to predict future energy [9-11].

1) Solar PV subsystem

Solar PV systems use the photovoltaic effect to turn sun radiation into power. Semiconductor materials with P-N junctions employ this method to generate energy when exposed to light [12

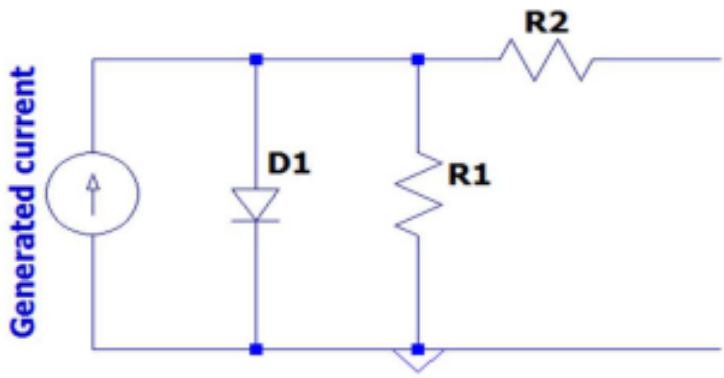


Fig. 1. solar cell's equivalent circuit

The generated current in the secondary side of each solar panel is obtained as follow:

$$I_o = I_{sh} - I_s\left(\exp\left(\frac{V+IR_2}{nV_t}\right) - 1\right) - \frac{V_o+IR_2}{R_1} \qquad (1)$$

Where Vo and Io are the generated voltage and current. I-V curves for solar cells show how current changes with voltage in a photovoltaic device. Characteristics of Solar Cells I-V Curves are essentially a visual summary of a module's operation that shows how current, and voltage relate to each other under the different irradiance and temperature circumstances [13]. Fig. 2 depicts the I-V, P-V curves of the proposed solar module.

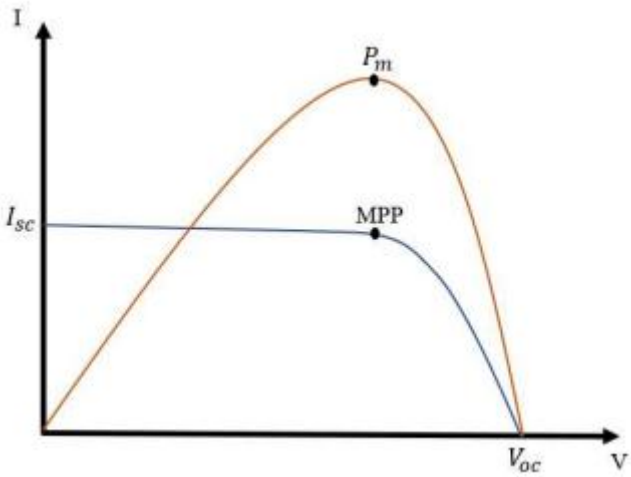


Fig. 2. Solar module I-V and P-V characteristic

An MPPT approach with a buck-boost converter is required in order to obtain the highest amount of energy from the sun. The P&O approach is used in this paper to track the maximum power points at a different time of the day. If the output power of the system is more than the needed energy for the charging station, as a demand side management with utilizing a multilevel inverter, the extra power can feed the power grid [14, 15].

2) Wind turbines subsystem

Recent years have seen the emergence of wind energy due to the quick development of new technology and wind turbines. A rotor, a set of blades, and a generator are the components of a wind turbine system that transform the energy of the wind into electric power. The amount of usable wind energy depends on both windspeed as well as the surface of the revolving blades [16, 17]. No need to mention that MPPT is necessary to maximize the use of wind energy. For various wind conditions, these MPPT algorithms monitor the maximum power. The goal of this strategy is to get the most power possible out of the wind turbine system by measuring the output power in real time and monitoring the rate at which power changes with speed. MPPT can be achieved by modifying the converter's duty cycle or rotor speed. The equation below is known as the tip speed ratio (TSR) [18].

$$\lambda = \frac{R\omega}{Velocity} \qquad (2)$$

where is the electromechanical frequency of the generator and velocity is the wind speed, gives the definition of tip speed ration which is very significant definition as seen in Fig. 3.

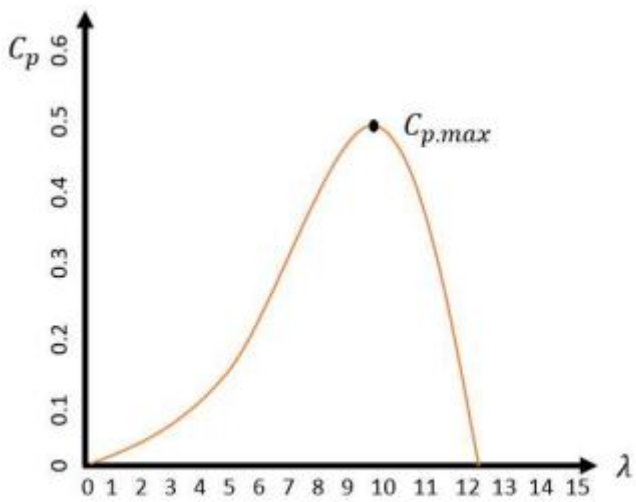


Fig. 3. Cp vs λ

Additionally, the wind turbine's maximum power may be obtained as follow:

$$P_{max} = \frac{1}{2} C_p \rho v^3 \qquad (3)$$

where) is the air density. According to Fig. 4 and the above equation, the highest amount of power "#$% is obtained with maximum value of ,-. and !,-.

III. PROPOSED EV CHARGING STATION

The proposed EV charging station in this paper contains two renewable energy sources including wind and solar system. The wind system contains 10 wind turbines and the nameplate capacity for the whole wind system is 600kW. Deimos DT2160 is utilized in this paper with three blades and the rotor diameter is 21 meters, the hub height of 50 meters, with the cut-in wind speed of 3 meter per second and cut-out wind speed of 25 meter per second. Fig. 4, shows the power curve of a turbine.

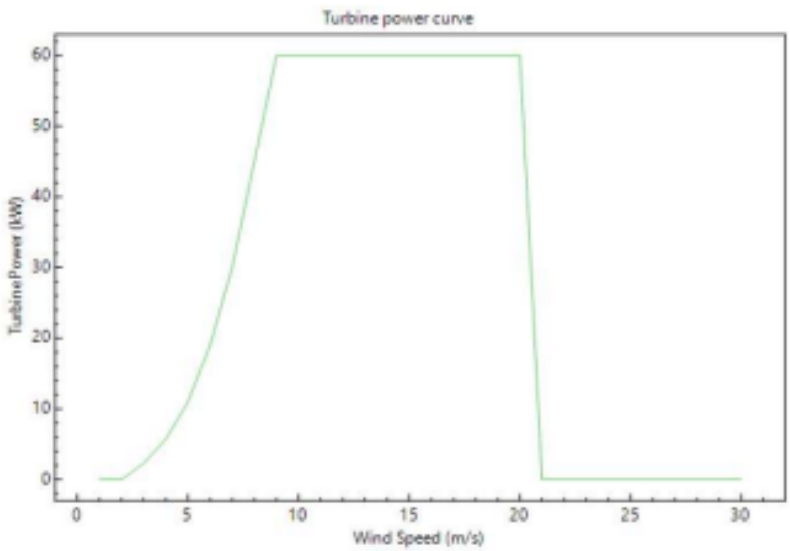


Fig. 4. Turbine power curve

As it can be seen, the output power of each turbine is 60 kW. In addition, the swept area by the blades is 362 square meters. Furthermore, table 1 shows the solar system used in this paper which has the nameplate capacity of 500kW.

TABLE 1. Presented solar system

Cells numbers per module	72	Efficiency	21.08 %
Module Size	1.6*1.14 m	Weight	18.5 kg
Rated power	339.3 W	Tilt	33.5 °
I (SC)	6.54 A	V(OC)	74.6 V
I (MPP)	6.14 A	V (MPP)	62.2 V

According to table 1, each module has 72 cells in series, and the weight of each module is 18.5 kg. the rated power is 339W and the tilt angle for the solar panels is 33.5 degrees which is equal to Lubbock latitude angle. Using solar and wind at the same time increase the reliability of the system. The proposed charging station block diagram is seen in Fig. 5.

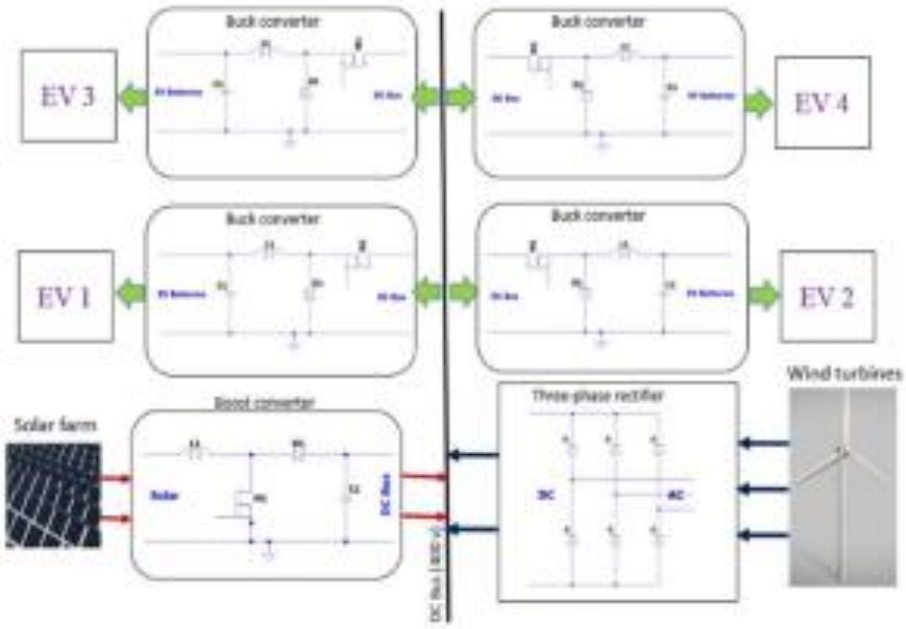


Fig. 5. Charging station block diagram

During the day, solar and wind energy are used to recharge the battery. By stepping up the voltage to 400V using a boost converter, the solar panel's output is managed. Since wind is present at night, it partially charges the DC bus. Then the EVs are charged by battery via utilizing a buck converter. Solar energy is widely available and cost-free, but power stability is challenging. However, it may be effectively made up for by fusing with another energy

source. As a result, in this study, solar and wind energy are combined. When there is wind, the battery is charged. When it is hot, a PV module may be used to charge the battery. This research proposes a hybrid system that is unusual in that it generates power from two separate renewable sources. When the hybrid solar/wind system is unable to absorb enough energy, the extra energy stored in the battery is utilized to charge EVs [20, 21].

CHAPTER-IX

SIMULATION RESULTS

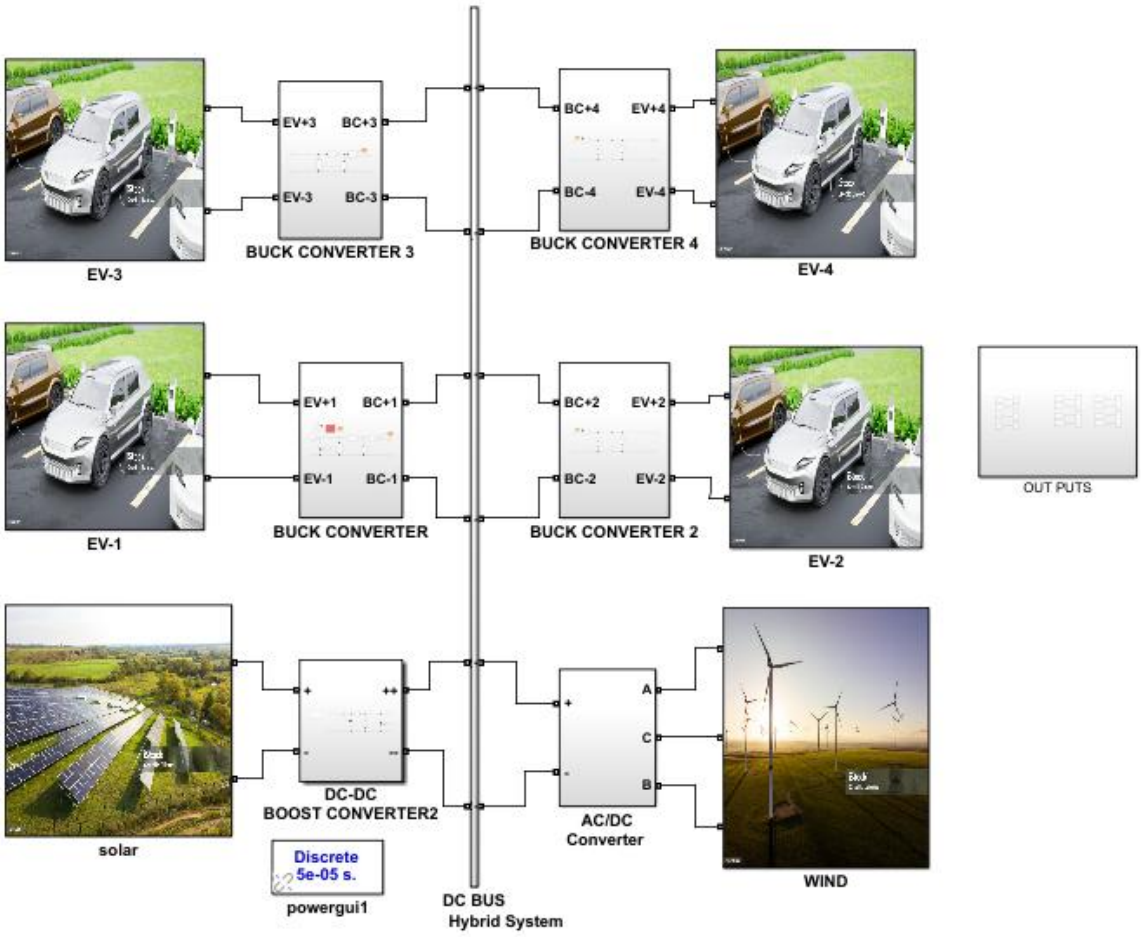


Fig 1 Simulation Diagram of proposed System.

Figure 1 illustrates the simulation diagram of the proposed system, which integrates a hybrid power generation system comprising both solar and wind energy sources for charging electric vehicles.

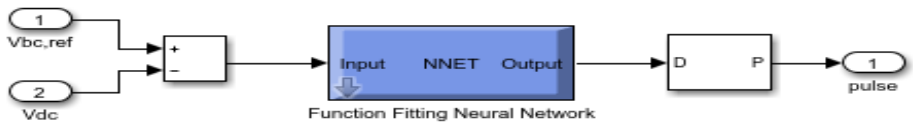


Fig 2 Buck converter Controller with ANN Control.

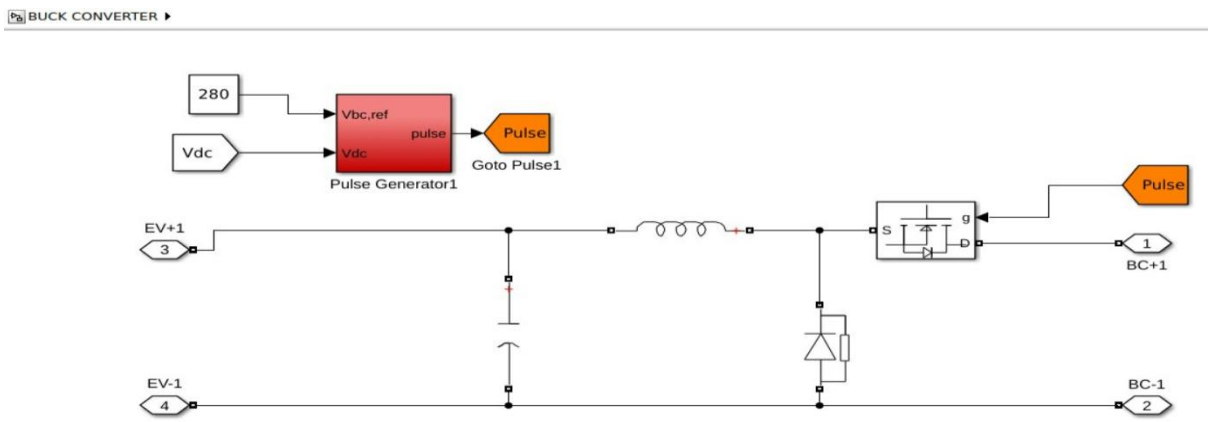


Fig.buck converter with ann controller in simulation

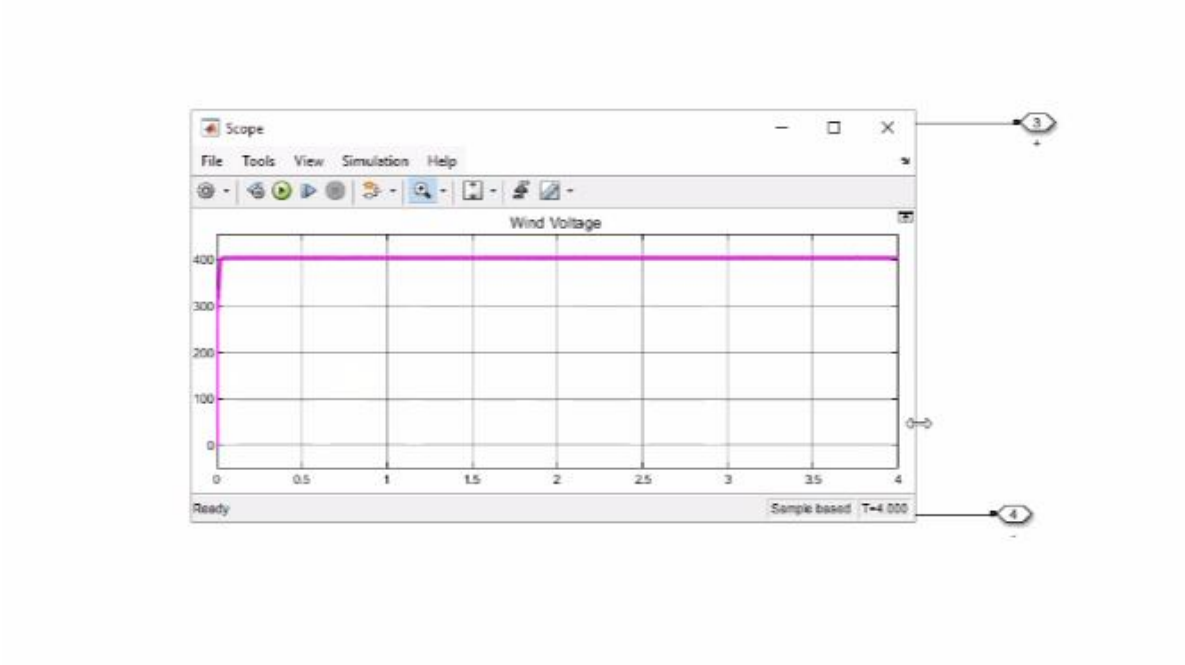
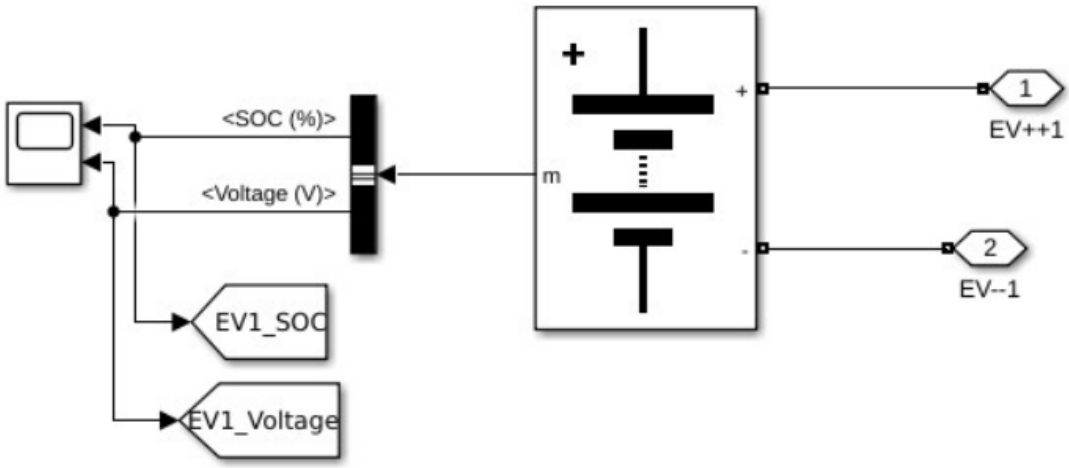
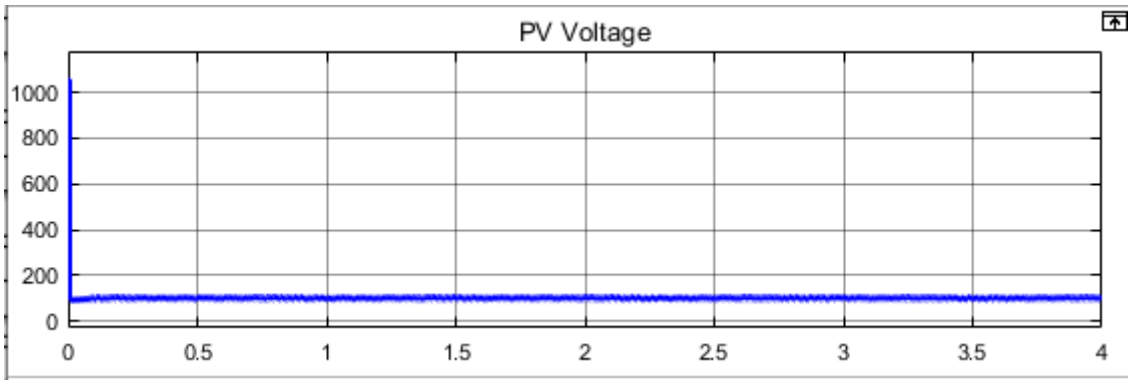


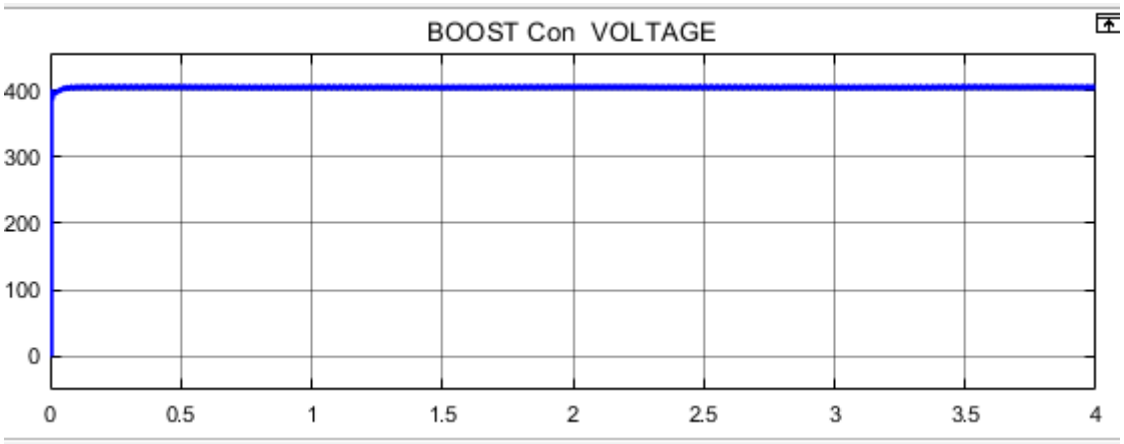
Fig.wind power

Fig .electric vehicle in simulation





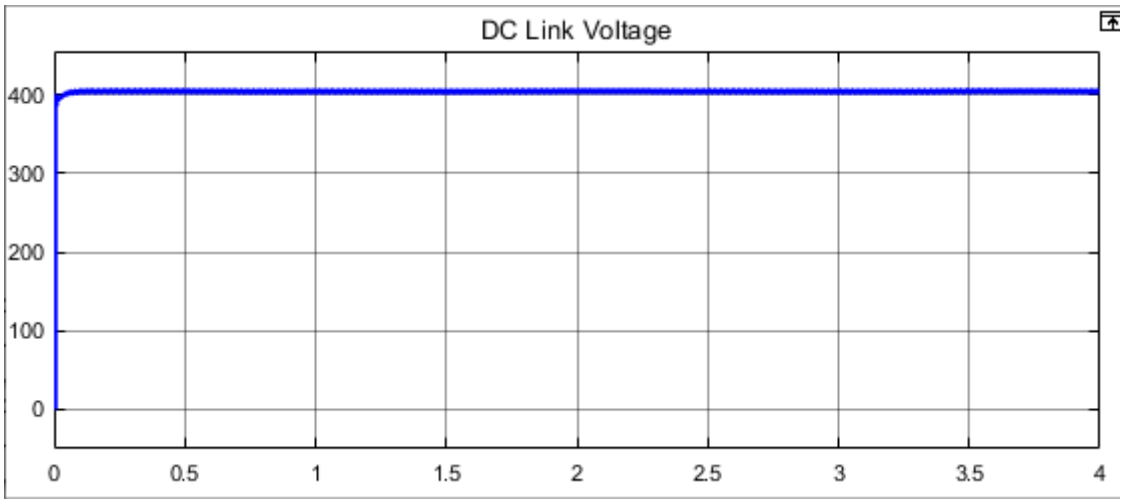
3(a) Solar Out Put



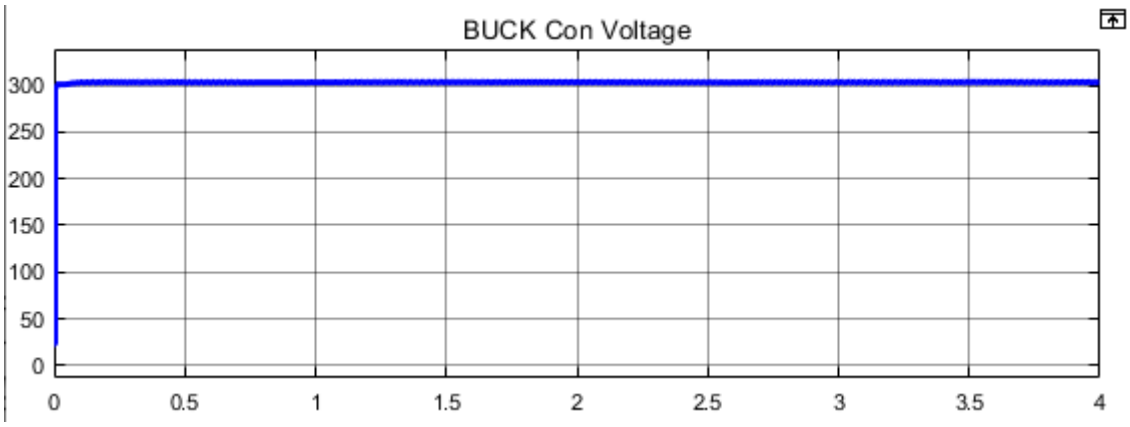
3(b) Boost converters Out Put

Fig. 3. Output voltage of the utilized boost converter and PV System

Fig. 3, the employed boost converter step up the voltage from 100V to 400V in the proposed DC fast charging station



4(a) DC link Voltage



4(b) Buck Converter Voltage

Fig. 4. Output voltage of the utilized buck converter And Dc link Voltage

Fig. 4 shows the output voltage of the buck converter which should be equal to the needed voltage of the EV batteries.

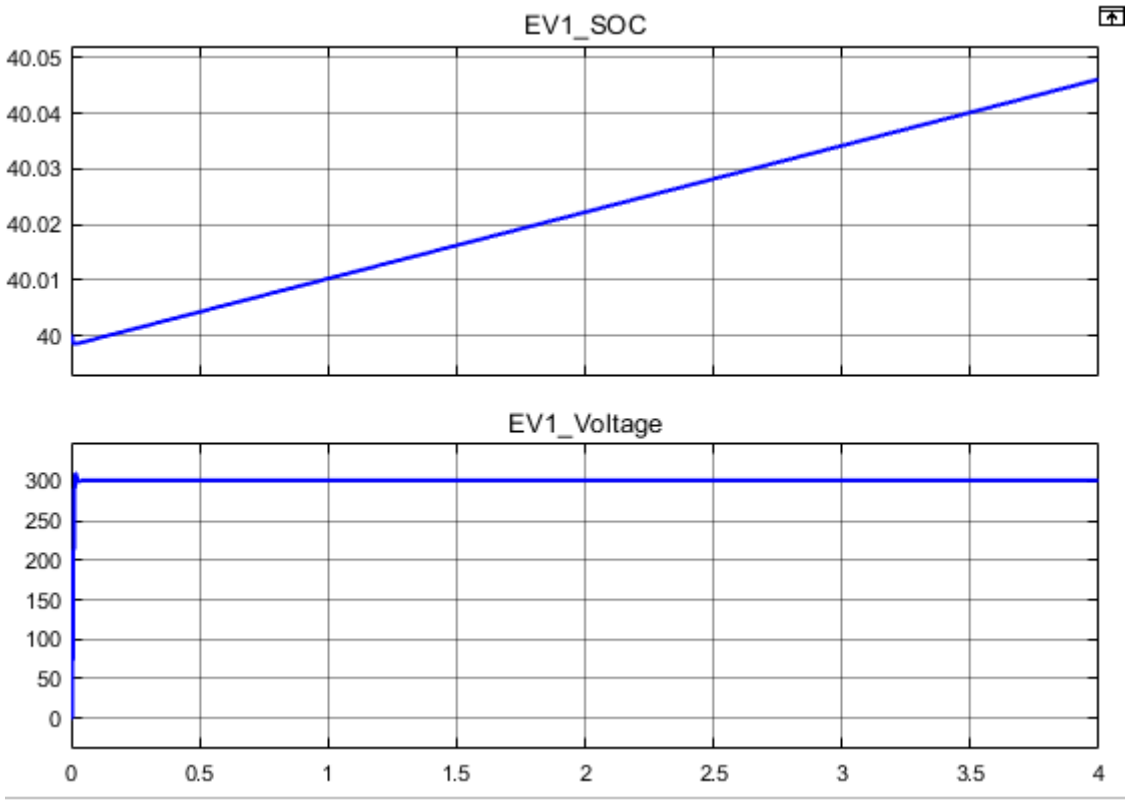


Fig5 EV1 Out Puts

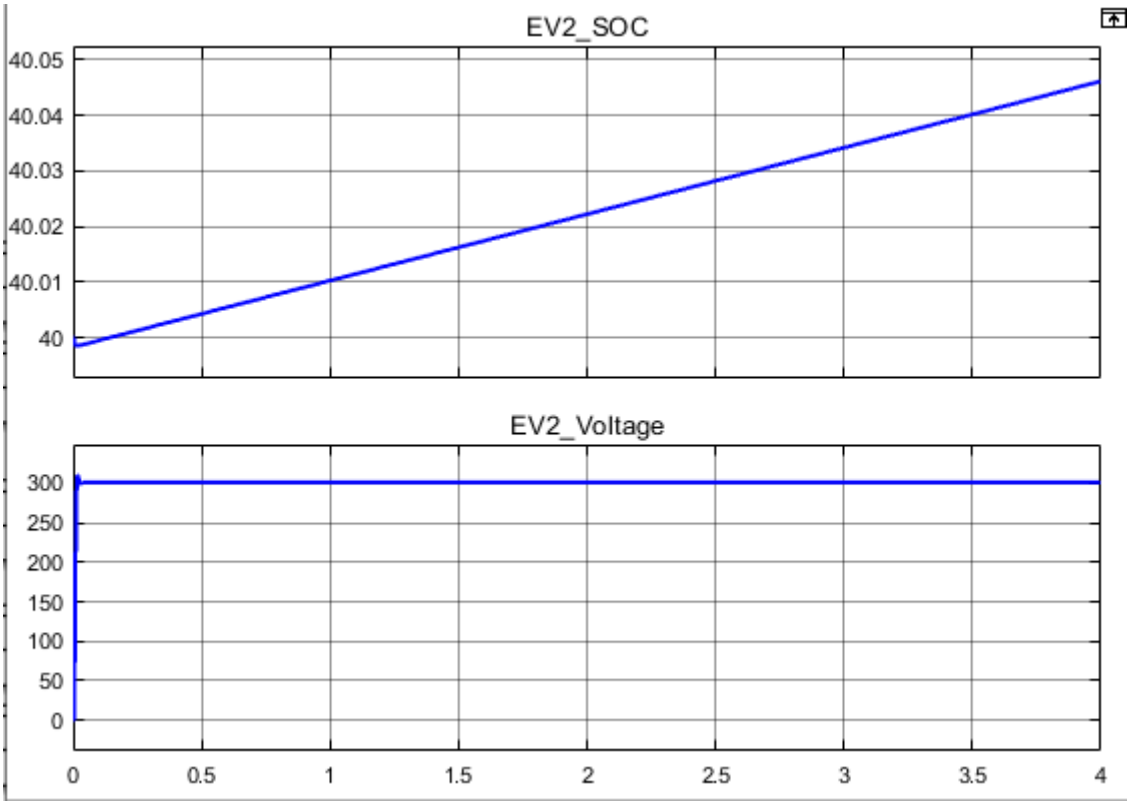


Fig6 EV2 Out Puts

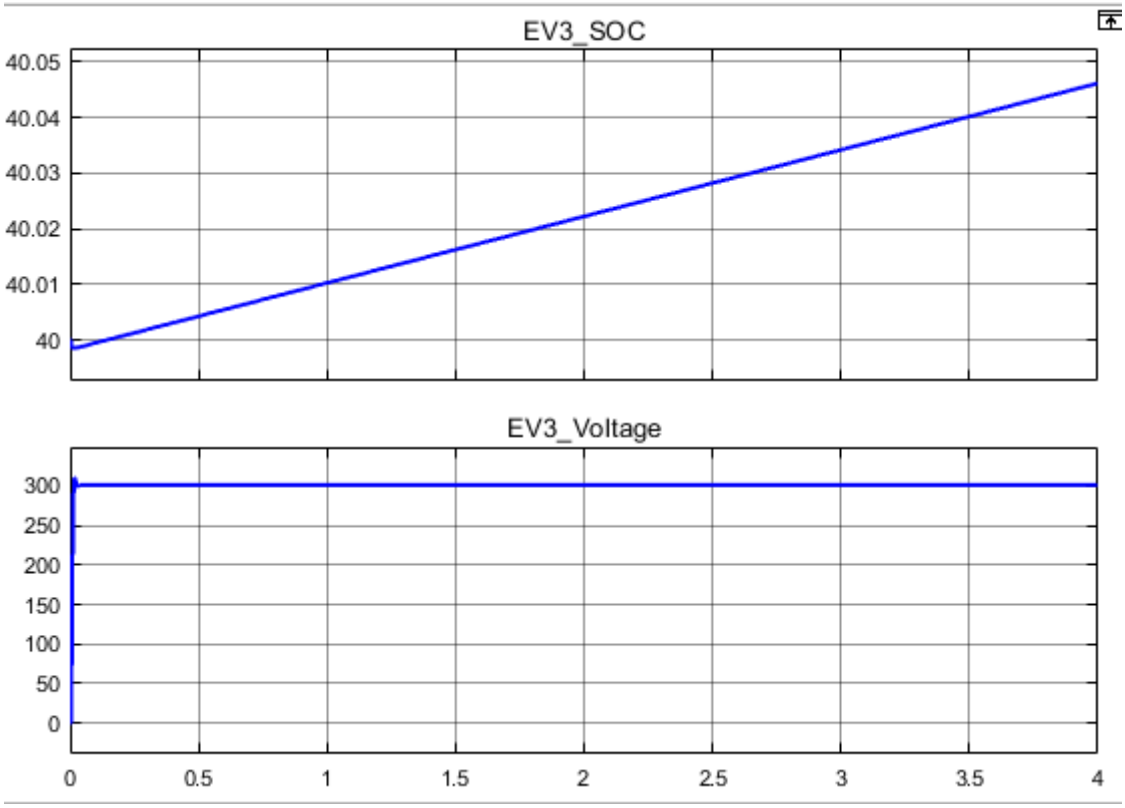


Fig7 EV3 Out Puts

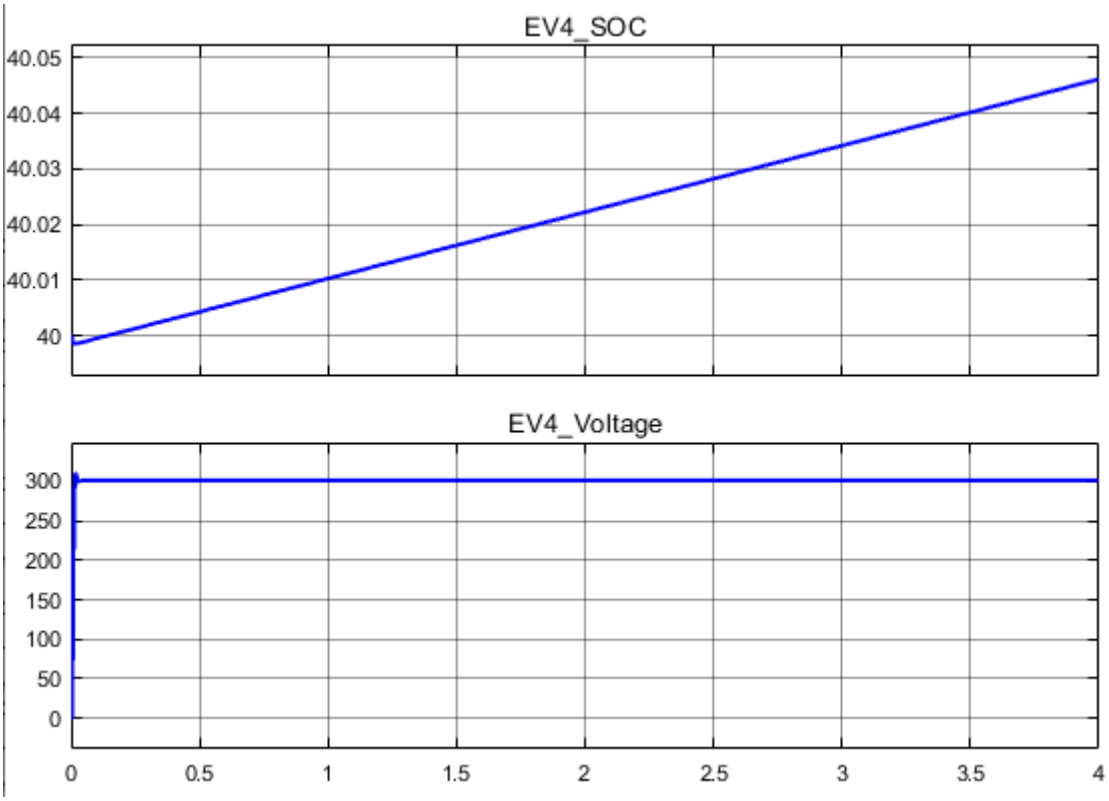


Fig 8 EV4 Out Puts

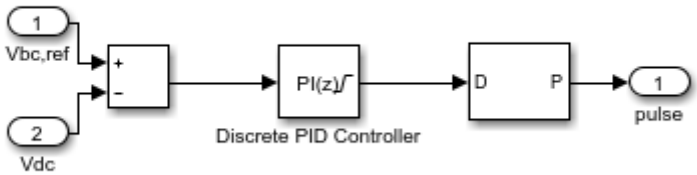


Fig 9 Buck converter Controller with PI Control.

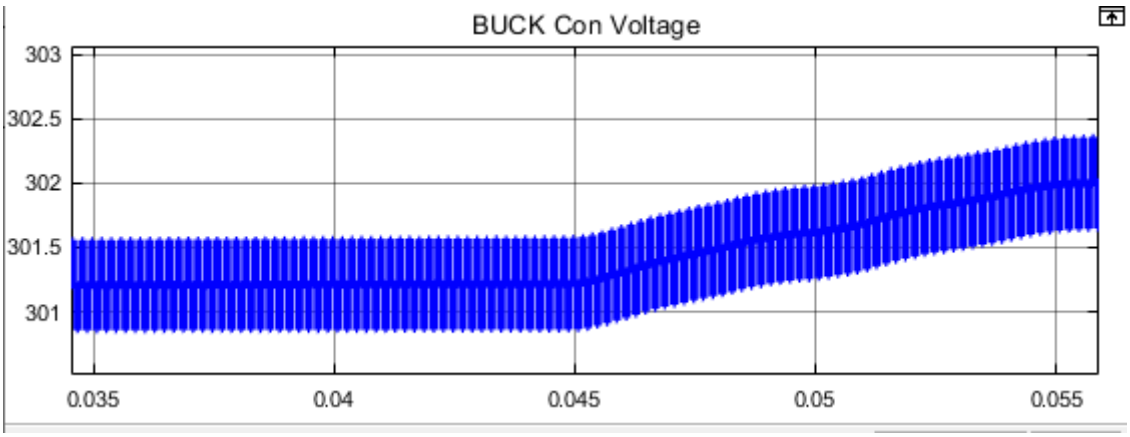


Fig 10 Zoomed results of Electrical Vehicle Battery Voltage with Buck converter Controlled by PI Controller

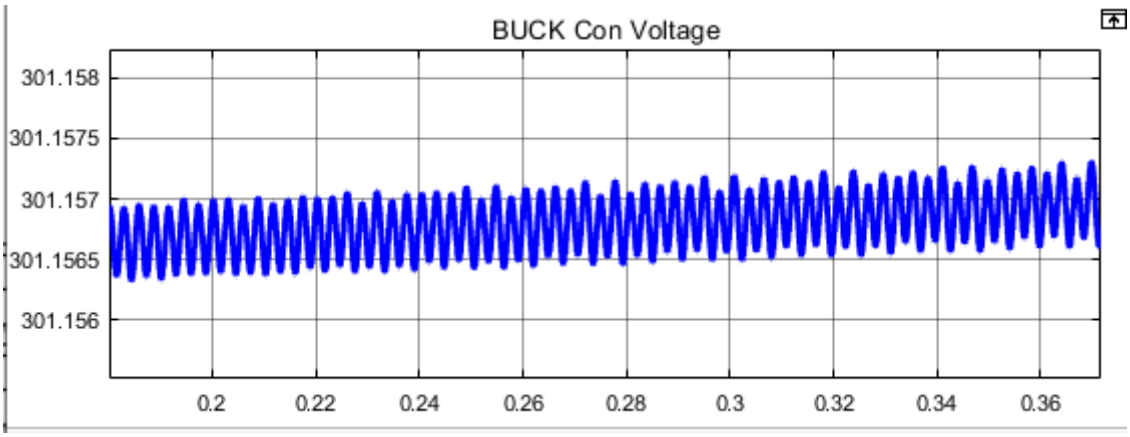


Fig 11 Zoomed results of Electrical Vehicle Battery Voltage with Buck converter Controlled by ANN Controller

Figures 10 and 11 illustrate the zoomed-in results of an Electric Vehicle (EV) battery voltage controlled by a Buck converter using both a PI Controller and an ANN Controller. Observations indicate that when the Buck converter is controlled by the PI Controller, the output voltage of the EV exhibits significant ripple content and more disturbed outputs. In contrast, utilizing the ANN Controller for the Buck converter results in a much smoother output voltage with minimal ripple. The reduced ripple content and smoother voltage levels achieved with the ANN Controller enhance battery performance and longevity. Excessive ripple and disturbed voltage outputs can lead to rapid battery degradation. Thus, the ANN Controller provides superior performance compared to the PI Controller, ensuring a more stable and reliable power supply for EV batteries.

CHAPTER-X

CONCLUSION

In conclusion, the design of a solar/wind hybrid power system for charging electric vehicles presents a sustainable and efficient solution to meet the growing demand for clean energy in transportation. This system leverages the complementary nature of solar and wind resources, ensuring a more reliable and continuous power supply. By integrating these renewable energy sources, the proposed hybrid system not only reduces dependence on fossil fuels but also minimizes environmental impact. This approach not only supports the transition to green energy but also provides a scalable and adaptable framework for future advancements in renewable energy technology and electric vehicle infrastructure.

REFERENCES

[1] H. S. Das, M. M. Rahman, S. Li, and C. Tan, "Electric vehicles standards, charging infrastructure, and impact on grid integration: A technological review," Renewable and Sustainable Energy Reviews, vol. 120, p. 109618, 2020.

[2] O. Ekren, C. H. Canbaz, and Ç. B. Güvel, "Sizing of a solar-wind hybrid electric vehicle charging station by using HOMER software," Journal of Cleaner Production, vol. 279, p. 123615, 2021.

[3] M. S. H. Lipu et al., "Review of electric vehicle converter configurations, control schemes and optimizations: Challenges and suggestions," Electronics, vol. 10, no. 4, p. 477, 2021.

[4] G. Kumar, "Optimal power point tracking of solar and wind energy in a hybrid wind solar energy system," International Journal of Energy and Environmental Engineering, vol. 13, no. 1, pp. 77-103, 2022.

[5] V. Khare, S. Nema, and P. Baredar, "Solar–wind hybrid renewable energy system: A review," Renewable and Sustainable Energy Reviews, vol. 58, pp. 23-33, 2016.

[6] L. Khalvati and L. Khalvati, "Estimating Potential Solar Energy With Three Different Architecture Designs Using Crystalline Silicon Modules."

[7] A. Balal and M. Giesselmann, "PV to Vehicle, PV to Grid, Vehicle to Grid, and Grid to Vehicle Micro Grid System Using Level Three Charging Station," in 2022 IEEE Green Technologies Conference (GreenTech), 2022: IEEE, pp. 25-30.

[8] Y. Wu, Z. Liu, J. Liu, H. Xiao, R. Liu, and L. Zhang, "Optimal battery capacity of grid-connected PV-battery systems considering battery degradation," Renewable Energy, vol. 181, pp. 10-23, 2022.

[9] A. T. Balal, M. Abedi, and F. Shahabi, "Optimized generated power of a solar PV system using an intelligent tracking technique," 2021.

[10] S. Ranjbari, T. Khatibi, A. Vosough Dizaji, H. Sajadi, M. Totonchi, and F. Ghaffari, "CNFE-SE: a novel approach combining complex network-based feature engineering and stacked ensemble to predict the success of intrauterine insemination and ranking the features," BMC Medical Informatics and Decision Making, vol. 21, no. 1, pp. 1-29, 2021.

[11] P. Pourmaleki, W. Agutu, A. Rezaei, and N. Pourmaleki, "Techno-Economic Analysis of a 12-kW Photovoltaic System Using an Efficient Multiple Linear Regression Model Prediction," International Journal of Robotics and Control Systems, vol. 2, no. 2, pp. 370-378, 2022.

[12] A. Balal and M. Giesselmann, "Demand side management and economic analysis using battery storage system (bss) and solar energy," in 2021 IEEE 4th international conference on power and energy applications (ICPEA), 2021: IEEE, pp. 141-146.

[13] G. Ciulla, V. L. Brano, V. Di Dio, and G. Cipriani, "A comparison of different one-diode models for the representation of I–V characteristic of a PV cell," Renewable and Sustainable Energy Reviews, vol. 32, pp. 684-696, 2014.

[14] A. Balal, S. Dinkhah, F. Shahabi, M. Herrera, and Y. L. Chuang, "A Review on Multilevel Inverter Topologies," Emerging Science Journal, vol. 6, no. 1, pp. 185-200, 2022.

[15] M. Alipour, J. Zarei, R. Razavi-Far, M. Saif, N. Mijatovic, and T. Dragičević, "Observer-based backstepping sliding mode control design for microgrids feeding a constant power load," IEEE Transactions on Industrial Electronics, vol. 70, no. 1, pp. 465-473, 2022.

[16] N. Sezer, Y. Biçer, and M. Koç, "Design and analysis of an integrated concentrated solar and wind energy system with storage," International Journal of Energy Research, vol. 43, no. 8, pp. 3263-3283, 2019.

[17] M. Nasser, T. F. Megahed, S. Ookawara, and H. Hassan, "Performance evaluation of PV panels/wind turbines hybrid system for green hydrogen generation and storage: Energy, exergy, economic, and enviroeconomic," Energy Conversion and Management, vol. 267, p. 115870, 2022.

[18] Y. Zhang, V. Ramdoss, Z. Saleem, X. Wang, G. Schepers, and C. Ferreira, "Effects of root Gurney flaps on the aerodynamic performance of a horizontal axis wind turbine," Energy, vol. 187, p. 115955, 2019.

[19] M. Bourhis, M. Pereira, F. Ravelet, and I. Dobrev, "Innovative design method and experimental investigation of a small-scale and very low tip-speed ratio wind turbine," Experimental Thermal and Fluid Science, vol. 130, p. 110504, 2022.

[20] J. Nishanthy, S. Charles Raja, T. Praveen, J. Jeslin Drusila Nesamalar, and P. Venkatesh, "Techno-economic analysis of a hybrid solar wind electric vehicle charging station in highway roads," International Journal of Energy Research, vol. 46, no. 6, pp. 7883-7903, 2022.

[21] L. Erickson and S. Ma, "Solar-powered charging networks for electric vehicles," Energies, vol. 14, no. 4, p. 966, 2021.